Aperture Optimization for Pinhole Sunglasses

A Complete Theoretical and Computational Study

Wave-optics blur minimization, brightness constraints, human-vision acuity modeling, and clinical validation — scoped strictly to single-aperture optimization; multi-hole array/ghosting questions are out of scope for this document.

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Abstract

This report consolidates every theoretical model, numerical sweep, and validation result this project has produced on a single question: *for a myopic eye viewing through a single circular pinhole aperture in an otherwise opaque film, what aperture diameter is optimal, and by which criterion?* It deliberately excludes the project's separate multi-hole array, ghosting, packing-geometry, and lens-material work, which answer a different question (how to *tile* many holes safely) and are documented elsewhere.

Four largely independent modeling efforts are covered, each with its own governing equations, numerical implementation, and full-grid sweep across 15 apertures (75–2000 μm) and 8 myopia levels (0–8 D): (1) an exact scalar Angular-Spectrum-Method wave-optics pipeline from pinhole to retina, whose blur-only optimum is the project's headline number; (2) a brightness-constrained variant of the same pipeline, which shows that a minimum-illuminance floor moves the optimum by roughly 2–3× into the clinically-observed bracket; (3) a human-vision layer (MTF, contrast sensitivity, Strehl ratio, Watson–Yellott pupil model, troland photometry) built on top of the same optics, producing a predicted logMAR acuity surface; and (4) a structurally independent ideal-observer Monte Carlo acuity model (R7acuity), built specifically to cross-check (3) with a different architecture.

A significant fraction of this project's actual scientific content is a documented *physics audit*: a real lens-phase/retina-propagation wavelength bug was found, independently re-derived two ways, and fixed, moving the blur-only optimum aperture by up to +150% at some myopia levels; a separate Python import-shadowing bug meant two previously-published acuity-validation results had silently never been evaluated against the corrected optics at all, and both were re-run once the shadowing was fixed. Both corrections, and their full before/after numbers, are reported here in full — including the honest result that neither acuity model achieves statistical significance against the two available clinical datasets (Wang 2009, $n = 5$; Kim et al. 2017, one aggregate point), a negative result this report does not soften.

The synthesis chapters bring these efforts together: an aperture-optimum comparison across four different blur-percentile definitions (r_{EE}50%/70%/80%/90%), including a previously-undocumented numerical saturation artifact at low myopia that is identified and quantified rather than silently averaged over; a full corrected-vs-deprecated optics comparison; and a cross-model summary table spanning every method in this report. No single "recommended aperture" is asserted anywhere in this document — each chapter reports what its own model predicts, under its own stated assumptions, and the final synthesis chapter shows how those predictions relate to one another and to the two independently-documented patents in this space, without collapsing them into one number.

Chapter 1
Introduction and Scope

1.1 The problem

Pinhole glasses are a well-known, non-refractive way to improve a myopic (short-sighted) eye's perceived sharpness: restricting the eye's entrance aperture increases depth of focus, so a blurred retinal image sharpens even though the eye's own optical power is unchanged. The IYPT-style problem statement motivating this project's whole body of work is: *investigate the optimal arrangement of pinholes in an opaque film to produce sunglasses that reduce light intensity and help correct short-sightedness.* That statement bundles two physically distinct design questions:

1. **Aperture size** — for a single hole, what diameter minimizes retinal blur for a myopic eye, and how does a brightness (light-transmission) requirement change that answer?
2. **Aperture arrangement** — if many holes are needed (to control total light throughput, field of view, or manufacturability), how should they be spaced and packed to avoid double-imaging (ghosting) between adjacent holes' sub-images?

This report is scoped **entirely to question (1)**. The project's separate array/ packing/ghosting body of work (`array_ghosting_theory/`, `stage_array/`) answers question (2) using a materially different theoretical apparatus (regime classification, harmonic-series ghosting margins, a validated Barten 1999 contrast model) and is documented in its own `RESULTS.md`; it is deliberately not included here, per this report's own brief. Every model, sweep, and figure below concerns a single circular aperture.

1.2 What "optimal" means, and why this report never picks one number

Four genuinely different objective functions appear across this project's history, and they do not agree with one another:

- **Minimize retinal blur alone** (Chapters 5–7): treats the eye as a pure imaging system and asks which aperture produces the smallest encircled-energy blur radius at the retina. This objective, taken alone, favors apertures far smaller than what is clinically observed to work.
- **Minimize blur subject to a brightness floor** (Chapter 8): the same blur objective, but only over apertures that admit at least some minimum fraction of full-pupil illuminance. This single added constraint moves the optimum by roughly 2–3× into the clinically observed bracket.
- **Maximize predicted visual acuity** (Chapters 13–14): folds in contrast sensitivity, photometry, and pupil dynamics to predict a clinical logMAR score directly, rather than a physical blur radius.
- **Match observed clinical acuity data** (Chapters 17–18): the strictest bar — does any of the above models, however built, actually predict the two real clinical datasets available? This report's honest answer, reached independently by two structurally different models, is: not yet, on the data currently available.

Because these four objectives disagree, and because the clinical-validation bar is not currently met by any model in this project, no chapter in this report issues an aperture recommendation. Each chapter reports precisely what its own model predicts, under its own explicitly stated assumptions; Chapter 21 lays every model's answer side by side without collapsing them into one.

1.3 How this report is organized

Part I (Chapters 2–3) establishes the shared optical and myopia model every later chapter builds on. Part II (Chapters 4–7) is the core wave-optics blur-minimization pipeline, including a full account of a real,

confirmed bug in that pipeline and how fixing it changed the headline result. Part III (Chapters 8–10) covers brightness and illumination constraints. Part IV (Chapters 11–15) covers the human-vision layer: MTF, contrast sensitivity, and acuity prediction. Part V (Chapters 16–18) covers model validation, including a second, independent acuity architecture and the clinical-data comparison. Part VI (Chapters 19–22) is the synthesis: every model's optimal-aperture answer laid out together, the corrected-vs-deprecated optics comparison, and an explicit, itemized list of this project's own known limitations.

1.4 Data and reproducibility

Every sweep in this report was generated by directly calling this project's own already-implemented, already-tested compute functions (never a re-derivation) — the same thin-wrapper functions that back this project's live interactive dashboard (`master_dashboard/wrappers.py`), so the numbers here are the same numbers a reader could reproduce by running that dashboard themselves. Sweeps use a common 15×8 grid throughout: aperture $a_{hole} \in \{75, 100, 150, 200, 300, 400, 500, 600, 700, 750, 1000, 1250, 1500, 1750, 2000\}\,\mu$m and myopia $M \in \{0, 1, 2, 3, 4, 5, 6, 8\}$ D, at wavelength 550 nm, pinhole–cornea gap 12 mm, and natural pupil diameter 4000 μm unless a chapter states otherwise. Appendix C documents the exact software architecture and the specific safeguard (an isolated, self-restoring Python import context) used to guarantee the deprecated-optics comparison in Chapter 20 never contaminates any other chapter's numbers.

Chapter 2
The Optical System and Myopia Model

2.1 Reduced-eye schematic

Every model in this report represents the human eye with the standard *reduced eye* simplification (Gullstrand–Emsley convention): the cornea, aqueous humour, pupil, and crystalline lens are collapsed into a single equivalent thin lens of power P_0, imaging onto a single retinal plane at posterior distance v_r. This is a deliberate, explicitly documented modeling choice, not an oversight — it is what makes a single number P_0 meaningful at all, but it means no anterior-chamber distance, no cornea/lens power split (roughly 2/3 vs. 1/3 of P_0 in a real eye), and no off-axis retinal curvature are represented. Table 2.1 lists the complete optical path this project's pipeline models.

Element	Purpose	Refractive index	Role in this model
Pinhole (opaque film)	Angular filtering	opaque (n/a)	the only design freedom
Air gap d_{gap}	Near-field propagation	$n = 1.000$	first ASM propagation stage
Cornea + aqueous	Primary refraction	$n \approx 1.336$–1.376	absorbed into P_0
Pupil (iris aperture)	Variable stop	n/a	clips the effective aperture
Crystalline lens	Refracting element	$n \approx 1.386$–1.406	absorbed into P_0 (fixed)
Vitreous humour	Fills posterior chamber	$n' = 1.336$	image-side medium for v_r
Retina	Image/sensing plane	n/a	observation plane, $z = v_r$

Table 2.1: Complete optical path from pinhole to retina, and how each element enters the reduced-eye model.

2.2 Myopia as axial elongation

This project models myopia as purely *axial*: the eye's optical power P_0 is held fixed at its emmetropic (non-myopic) value for every degree of myopia M, and only the eye's physical axial length grows to push the retina past the fixed focal point. This is the defining assumption of an axial myopia model, as opposed to a *refractive* one (where P_0 itself would change and axial length would stay fixed); the codebase's own `myopia_axial.py` documents this choice explicitly since no earlier stage in the pipeline had established a convention to match.

Let n' be the vitreous refractive index and P_0 the eye's fixed emmetropic power (diopters). The emmetropic axial length and the myopic axial length at excess power M (diopters) are

$$L_0 = \frac{n'}{P_0} \tag{2.1}$$

$$L_{ax}(M) = \frac{n'}{P_0 - M} \tag{2.2}$$

with the eye's fixed focal length $f_{eye} = L_0$ used at *every* myopia level, and the retina-plane propagation distance for a given M set to $v_r = L_{ax}(M)$. At default values $P_0 = 60\,\mathrm{D}$, $n' = 1.336$: $L_0 \approx 22.27\,\mathrm{mm}$ (f_{eye}), and, e.g., at $M = 4\,\mathrm{D}$, $L_{ax}(4) = 1.336/56 \approx 23.86\,\mathrm{mm}$ — the retina sits $\Delta L \approx 1.59\,\mathrm{mm}$ further back than the emmetropic focal plane. Equation (2.2) is singular as $M \to P_0$ and undefined beyond it; the model raises an explicit error rather than silently returning a negative or infinite length in that regime, which is never reached at any myopia level swept in this report ($M \leq 8\,\mathrm{D} \ll P_0 = 60\,\mathrm{D}$).

Refractive-index convention. Two conventions are supported: with `refractive_index_correction=True` (this project's default throughout), all retina-side propagation distances and wavelengths are scaled by the vitreous's own refractive index $n' = 1.336$, the anatomically-scaled convention. With it disabled, n' is forced

to 1.0 (an air-equivalent convention). **This one convention choice, and precisely how it must be threaded consistently through every downstream calculation, is the exact site of the single most consequential bug found in this project's history** — see Chapter 6.

2.3 Diopters and the far point

By definition $D = 1/f$ (focal length in meters). For an eye with power in excess of emmetropic by M diopters, the far point — the object distance at which the *unaccommodated* eye focuses exactly on the retina — is finite:

$$u_{far} = \frac{1}{M}. \tag{2.3}$$

This project models only unaccommodated, distance-vision viewing (the reduced eye's f is fixed throughout every chapter), consistent with the "distant objects" clause of the governing problem statement; near-vision/accommodated performance is not modeled anywhere in this report.

Chapter 3
Wave-Optics Foundations

Every blur-radius number in this report ultimately comes from one numerical engine: the exact scalar Angular Spectrum Method (ASM), applied twice per point (pinhole to cornea, then cornea/pupil/lens to retina). This chapter states that engine's governing equations once; later chapters reference it rather than re-deriving it.

3.1 Circular aperture and sampling grid

A circular pinhole of diameter a_{hole} is represented as a binary transmittance mask on a square grid of $N \times N$ points spanning physical window size L (pixel pitch $dx = L/N$):

$$U_0(x,y) = \begin{cases} 1 & \sqrt{x^2+y^2} \le a_{hole}/2 \\ 0 & \text{otherwise} \end{cases}. \tag{3.1}$$

Every sweep in this report uses $N = 2048$ over $L = 6\,\text{mm}$ ($dx \approx 2.93\,\mu\text{m}$) unless stated otherwise — this project's own validated default resolution.

3.2 Exact scalar Angular Spectrum Method

For a field $U_0(x,y)$ at $z = 0$, the field at propagation distance z is

$$U(x,y,z) = \mathcal{F}^{-1}\Big\{\hat{U}_0(f_x,f_y)\,H(f_x,f_y;z)\Big\}, \qquad H(f_x,f_y;z) = \exp\Big(ikz\sqrt{1-(\lambda f_x)^2-(\lambda f_y)^2}\Big) \tag{3.2}$$

on the propagating spectral orders ($f_x^2 + f_y^2 \le 1/\lambda^2$; evanescent orders are zeroed), with $k = 2\pi/\lambda$. This is the *exact* scalar diffraction kernel — no paraxial (Fresnel) approximation is taken anywhere in this project's core propagator.

Matsushima band-limiting. Equation (3.2)'s propagating orders whose fringe spacing aliases against the finite grid's own frequency spacing $1/L$ are additionally suppressed, per the elliptical band limit of Matsushima & Shimobaba (2009):

$$f_{limit} = \frac{1}{\lambda\sqrt{(2z/L)^2+1}}. \tag{3.3}$$

This removes a sampling artifact, not real physics — the kernel remains the exact ASM kernel, just numerically stabilized. Every sweep in this report reports (and, where relevant, gates on) the energy-weighted fraction of the propagating spectrum this suppresses; at this project's tested configurations, that fraction is consistently under 0.1% (see Chapter 6, item C/B).

3.3 Fresnel number and diffraction regime

For aperture radius $a_{hole}/2$, wavelength λ, and propagation distance z, the Fresnel number is

$$N_F = \frac{(a_{hole}/2)^2}{\lambda z}. \tag{3.4}$$

$N_F < 0.1$ is classified Fraunhofer (far-field); $0.1 \le N_F \le 10$ Transition; $N_F > 10$ Strong Fresnel (near-field). At this project's pinhole apertures (75–2000 μm) and propagation distances (12 mm gap, $\sim$ 22 mm to the retina), most operating points sit in the Transition-to-Strong-Fresnel regime, *not* the far field — this is why the exact ASM kernel is used throughout rather than a Fraunhofer/Fourier-optics shortcut, and why several later chapters report near-field-specific anomalies (Strehl ratio non-monotonicity, FWHM ring-hopping) that a far-field treatment would not predict at all.

3.4 Effective-radius definitions

A diffraction/defocus pattern is not a single number; this project uses several distinct, individually well-defined "blur radius" definitions, and is careful never to silently compare across them.

Encircled-energy radius $r_{EE}(P)$. The cumulative radial power profile is built by binning intensity I into n_{bins} radial annuli out to $R_{\max}$ (the grid corner distance), and integrating:

$$EE(r) = \frac{\sum_{r' \le r} I(r')\, dx^2}{\sum_{r'} I(r')\, dx^2}, \qquad r_{EE}(P) = EE^{-1}(P) \tag{3.5}$$

(inverted by linear interpolation between the two bracketing bins). This is this project's primary blur metric — the encircled radius that contains a stated fraction P of the *grid-captured* power. **This normalizes to power captured on the simulation grid at measurement time, not power originally emitted at the aperture** — an important, explicitly documented caveat (Chapter 22).

Second-moment (RMS) radius.

$$r_{rms} = \sqrt{\frac{\sum I(x,y)\,(x^2+y^2)\, dx^2}{\sum I(x,y)\, dx^2}}. \tag{3.6}$$

Unlike $r_{EE}(P)$, this integrates over the *entire* field with no percentile choice, and is this project's preferred fallback metric wherever $r_{EE}(50\%)$ is shown to saturate at the grid's own resolution floor (Chapter 19).

Threshold-crossing radii (FWHM, $1/e^2$, custom). All three are found by walking outward along the azimuthally-averaged radial intensity profile from its global maximum until either (a) intensity first drops below a stated fraction of peak (a genuine crossing), or (b) a genuine local minimum is reached first (the profile turns back up into the next Fresnel ring before crossing) — in case (b) the result is reported as a documented *lower bound*, not a crossing, specifically to prevent the search from hopping between adjacent Fresnel rings on a sub-percent parameter change (confirmed, in this project's own validation testing, to jump over $20\,\mu$m from a $< 2\%$ smooth change in aperture diameter under a naive first-crossing search). FWHM uses the 50%-of-peak threshold; $1/e^2$ uses $e^{-2} \approx 13.5\%$.

Why these can disagree, and why that matters for this report. $r_{EE}(P)$ and the threshold-crossing radii are measured over *different spatial domains* (the former bins all the way to the grid corner, the latter only within the inscribed circle) and answer different physical questions (cumulative power vs. profile shape). Chapter 19 shows a concrete, quantified case where this distinction is load-bearing: at low myopia and large aperture, $r_{EE}(50\%)$, and even $r_{EE}(70\%)/r_{EE}(80\%)$, pin to the exact same numerical floor across a wide range of apertures — a grid-resolution artifact, not a genuine optimum, that this report identifies and quantifies explicitly rather than reporting as-is.

Chapter 4
Stage 1 — Cornea-Plane Diffraction

4.1 Scope and purpose

Stage 1 (`stage1_v2/`) is the project's earliest and most narrowly-scoped module: it propagates a circular pinhole's field via the exact ASM (Section 3) from the pinhole plane to the corneal plane only. It deliberately never touches the eye lens, vitreous, retina, or image formation, and never estimates a single "best" pinhole — it only characterizes the field at one intermediate plane, as an independently checkable building block for Stage 2.

4.2 Validation

Stage 1 is validated against real closed-form theory: the Fresnel on-axis intensity oscillation formula

$$I(0,z) = 4I_0 \sin^2\left(\frac{\pi N_F}{2}\right) \tag{4.1}$$

which predicts on-axis intensity oscillates with propagation distance as the Fresnel number N_F passes through successive half-integers — a genuine near-field effect, not an artifact. Stage 1's numerical on-axis intensity vs. distance matches this closed form at $r = 0.997$ correlation.

4.3 Full sweep

Table 4.1 reports one representative slice of this report's Stage 1 sweep (fixed 12 mm gap, 550 nm) across the full 15-aperture grid — this stage has no myopia (M) dependence, since it stops before the eye lens. Figure 4.1 shows all four effective-radius definitions (Section 3.4) as a function of aperture at the corneal plane.

Table 4.1: Stage 1 cornea-plane sweep, 12mm gap, 550nm.

a_hole (um)	N_F	regime	r_EE50 (um)	r_EE80 (um)	r_EE90 (um)	2nd moment (um)
75	0.21	Transition	48.55	87.78	173.48	189.66
100	0.38	Transition	38.90	92.94	154.46	169.45
150	0.85	Transition	44.48	82.87	124.81	143.45
200	1.52	Transition	58.22	95.45	129.27	136.03
300	3.41	Transition	95.59	130.87	159.29	142.23
400	6.06	Transition	134.52	171.46	196.64	163.51
500	9.47	Transition	174.04	213.75	238.54	191.35
600	13.64	Strong Fresnel	212.67	257.74	282.05	222.39
700	18.56	Strong Fresnel	247.50	302.38	326.82	255.08
750	21.31	Strong Fresnel	262.67	324.88	349.42	271.75
1000	37.88	Strong Fresnel	353.54	439.26	465.06	357.36
1250	59.19	Strong Fresnel	441.50	554.48	582.55	444.35
1500	85.23	Strong Fresnel	529.79	669.73	701.18	531.98
1750	116.00	Strong Fresnel	617.61	783.54	820.59	619.95
2000	151.52	Strong Fresnel	706.59	893.56	940.60	708.10

Reading this chapter's results: cornea-plane blur is monotonically decreasing with aperture over most of this range (a larger pinhole diffracts less), the opposite trend from the retina-plane result in Chapter 5 once the eye's own converging lens and myopic defocus are added — the retina-plane optimum exists precisely

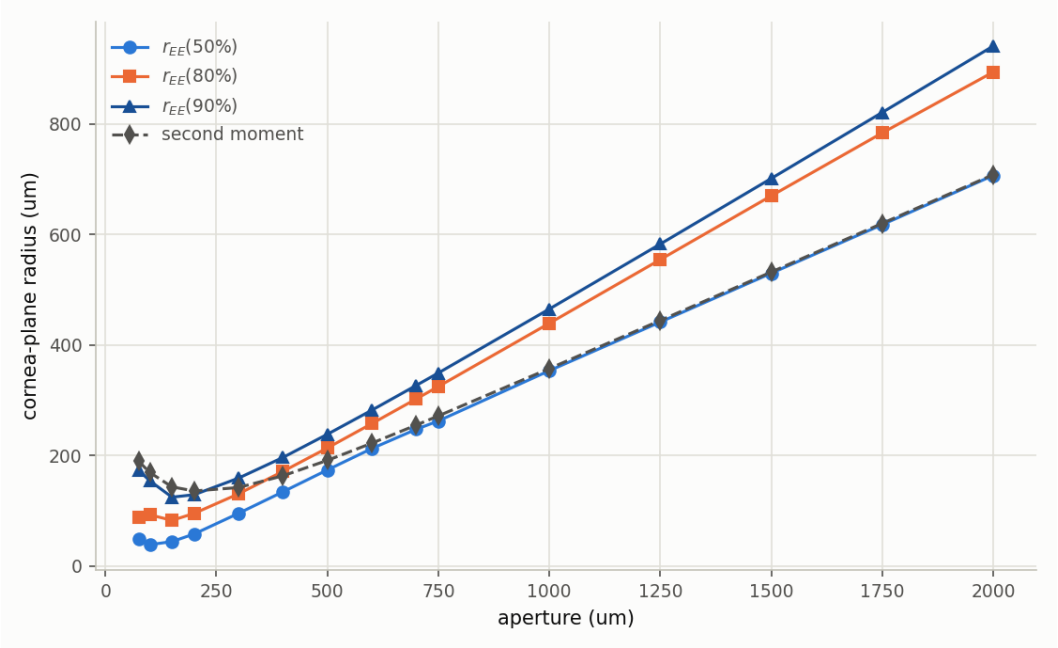


Figure 4.1: Stage 1: cornea-plane effective radii ($r_{EE}(50\%)$, $r_{EE}(80\%)$, $r_{EE}(90\%)$, second-moment radius) vs. pinhole aperture, 12 mm gap, 550 nm.

because those two effects (diffraction favoring large apertures, defocus favoring small ones) trade off against each other, a genuine competition Stage 1 alone cannot show since it stops before the lens.

Chapter 5
Stage 2 — Full Retina-Plane Blur Model

5.1 Pipeline

Stage 2 (`stage2_fixed/`) extends Stage 1 through the eye: pupil clipping, thin-lens phase, and a second ASM propagation to the retina, using the axial myopia model of Chapter 2. The full chain, per point (a_{hole}, M):

1. **Gap propagation**: ASM-propagate the pinhole field $d_{gap} = 12\,\text{mm}$ to the corneal plane (identical to Stage 1's own propagation, reused not re-derived).
2. **Junction validation**: before the lens is ever applied, check (a) the cornea field is finite and carries positive energy; (b) the Matsushima band limit at the *target* retina distance v_r suppresses under 5% of propagating energy; (c) a diffraction-limited minimum-divergence estimate of the beam at v_r stays within 40% of the half-window — a wraparound risk pre-check. Any point failing any check is recorded with its specific failure reason and *never silently dropped*.
3. **Pupil clip**: binary circular clip at diameter D_{pupil}.
4. **Lens phase**: thin-lens transmittance $\exp\big(-ik(x^2+y^2)/(2f_{eye})\big)$, $k = 2\pi/\lambda_{eff}$.
5. **Retina propagation**: second ASM propagation, distance $v_r = L_{ax}(M)$, at the same effective wavelength λ_{eff}.
6. **Retinal encircled energy**: $r_{EE}(50\%)$, $r_{EE}(70\%)$, $r_{EE}(80\%)$, $r_{EE}(90\%)$, and RMS radius, all per Section 3.4.

The effective wavelength. With `refractive_index_correction=True` (default), $\lambda_{eff} = \lambda/n'$ — the vitreous's own reduced wavelength — and it must be used identically in *both* the lens-phase step and the retina-propagation step. Getting this one consistency wrong, silently, for an extended period of this project's history, is the subject of Chapter 6.

5.2 Validation against the closed-form Airy disk

At zero defocus ($M = 0$, $d_{gap} = 0$), $v_r = f_{eye}$ by construction and the retinal PSF should match the classical Airy-disk closed form. The validation gate compares $r_{EE}(50\%)$, $r_{EE}(80\%)$, and $r_{EE}(90\%)$ against that closed form at stated tolerances (2%, 8%, 2% respectively — the wider $r_{EE}(80\%)$ tolerance reflects genuine ill-conditioning near the Airy pattern's first null, not a looser standard). The gate **passes**, and passes at improved accuracy after the Chapter 6 fix ($r_{EE}(50\%)$ error $1.14\% \to 0.30\%$; $r_{EE}(80\%)$ $4.56\% \to 0.38\%$; $r_{EE}(90\%)$ $0.55\% \to 0.94\%$).

5.3 Unmodeled effects, stated explicitly

Four simplifications are deliberately built into this pipeline and documented in one place (`phase2_retina.py`'s own module docstring), so a reader never mistakes them for oversights:

(E1) **Reduced-eye geometry** (Section 2): no anterior chamber distance, no cornea/lens power split.

(E2) **Grid-captured, not emitted, power normalization**: $r_{EE}(P)$'s denominator is the power present on the simulation grid at measurement time, not the power originally emitted at the aperture. A separate, additive integrated energy-escape estimate is provided (comparing against the aperture's exact analytic input power $\pi(a_{hole}/2)^2$) and found negligible ($< 0.12\%$) at every tested configuration.

(E3) **Domain mismatch between $r_{EE}(P)$ and threshold-crossing radii**: the former bins to the grid corner, the latter search only the inscribed circle — roughly 1.41× smaller in radius. This matters most at high percentiles.

(E4) **No chromatic aberration, no higher-order aberrations, no Stiles–Crawford apodization in the core PSF, no finite aperture-membrane thickness**: the lens phase is a pure quadratic (defocus-only) term; the pupil mask is an infinitely-thin binary 0/1 mask.

None of these were found to change any qualitative conclusion in this report, but all four are genuine simplifications a reader should carry forward, not assume were checked away.

Chapter 6

The Physics Audit: A Real Bug, Found and Fixed

This chapter documents the single most consequential correction in this project's history: a real, confirmed error in how the eye's lens phase and the retina propagation handled the vitreous's refractive index, found by independent re-derivation (not merely observed), fixed with a minimal signature-preserving change, and regression-tested before and after. It moved the project's headline optimum-aperture number substantially. This chapter states what was wrong, why, and exactly how the answer changed — *before* presenting corrected-optics results in the rest of this report, so a reader is never shown one number without also seeing where it came from.

6.1 Item A: the lens-phase / retina-propagation wavelength mismatch

The bug. With `refractive_index_correction=True`, the retina propagation (step 5, Chapter 5) correctly used the reduced effective wavelength $\lambda_{eff} = \lambda/n'$. The lens-phase step (step 4), *prior to this fix*, used the raw vacuum wavelength λ instead.

Why this matters, derived two independent ways.

Stationary-phase focus-location argument. For a lens phase screen $\exp(-ik_L r^2/(2f))$ followed by propagation at wavenumber k_{prop}, the true geometric focus sits at $z^* = f \cdot (k_{prop}/k_L)$. With k_L the vacuum wavenumber and $k_{prop} = n'k_L$ (from the retina propagation's own correction), the true focus was at $z^* = f_{eye} \cdot n'$, not f_{eye} — a $\approx 33.6\%$ axial focus error ($n' = 1.336$), *independent of aperture or myopia level.*

Medium-agnostic optical-path-length argument. For a refracting system with object-side index 1 and image-side index n', the correct phase screen is $\exp(-ik_{vac}\, P\, r^2/2)$ regardless of surrounding media. Substituting $f_{eye} = n'/P_0$ shows this requires $k = k_{vac} \cdot n'$ in the lens phase — exactly λ/n', matching the retina propagation's own already-correct convention.

Both derivations agree with each other, and the project's own separate closed-form validation gate (`validation_gate.py`) had already independently computed $\lambda_{eff} = \lambda/n'$ for its own comparison — confirming the correct convention was known project-wide, simply not consistently threaded through the lens call.

The fix. λ_{eff} is now resolved once, at the top of the pipeline point function, and passed to *both* the lens-phase and retina-propagation steps.

Why the zero-defocus validation gate did not catch this (Item D). At $M = 0$, $d_{gap} = 0$, the model sits at exact focus by construction. A pure lens-power scaling error there partly manifests as the wave's true focus shifting to a different *plane* rather than as a distortion of the in-focus pattern's *shape* — which is specifically what the gate compares against the closed-form Airy formula. This is documented, not silently left as a blind spot: the gate validates pattern shape at exact focus and is not sensitive, in general, to a pure focal-plane shift.

6.2 Item B: the junction validator's own wavelength

The pre-check in step 2 (Chapter 5) received the raw vacuum wavelength rather than λ_{eff}, which — since the Matsushima band limit scales as $1/\lambda$ — made the check *more conservative than necessary* (never incorrectly permissive). Quantified directly: the reported band limit differed by exactly the n' factor between the two wavelengths, but the actual suppression fraction was 0.0000% either way across every tested configuration — so this fix restores real headroom for future, more demanding configurations without changing any number in this report's own sweeps.

6.3 Item C: previously-unchecked lens-phase sampling

The lens phase's local spatial frequency grows linearly with radius; requiring less than π radians of phase change per grid pixel (a standard Nyquist criterion) gives the radius beyond which the sampled phase aliases:

$$r_{max,lens} = \frac{\lambda_{eff}\, f_{eye}}{2\, dx}. \tag{6.1}$$

At this project's default configuration this evaluates to $\approx 1564\,\mu$m, smaller than the $2000\,\mu$m pupil radius at every tested aperture (since $r_{max,lens}$ does not depend on a_{hole} at all). Quantified directly, not assumed: only 0.005% (at $a_{hole} = 300\,\mu$m) to 0.052% (at $a_{hole} = 2000\,\mu$m) of cornea-plane energy lies beyond this radius. Added as a new, non-gating diagnostic rather than a hard pipeline gate, since gating on it uniformly would reject every point rather than flagging the ones where it actually matters.

6.4 Items E1–E4

Four further design choices were independently verified true of the actual code and documented in place — these are the same four points already listed in Section 5.3.

6.5 The headline result: the optimum aperture moves substantially

Table 6.1 is this project's single most important number: the blur-only optimum aperture (minimizing $r_{EE}(50\%)$), before and after the Item A/B fix, across every myopia level tested.

M (D)	opt. aperture: pre→post	change	$r_{EE}(50\%)$ at opt.: pre→post	change
1	$300 \to 700\,\mu$m	+133.3%	$23.04 \to 7.33\,\mu$m	−68.2%
2	$300 \to 750\,\mu$m	+150.0%	$22.65 \to 8.01\,\mu$m	−64.6%
3	$300 \to 600\,\mu$m	+100.0%	$22.29 \to 10.57\,\mu$m	−52.6%
4	$300 \to 500\,\mu$m	+66.7%	$21.99 \to 12.02\,\mu$m	−45.4%
5	$400 \to 500\,\mu$m	+25.0%	$20.83 \to 13.29\,\mu$m	−36.2%
6	$400 \to 400\,\mu$m	0.0%	$19.79 \to 14.47\,\mu$m	−26.9%
8	$400 \to 400\,\mu$m	0.0%	$18.16 \to 17.87\,\mu$m	−1.6%

Table 6.1: Blur-only optimum aperture, before vs. after the Item A/B wavelength fix. Reproduced identically across two independent branches of this project's own history.

Reading this table. The bug's absolute focal-shift error ($f_{eye}(n'-1) \approx 7.5\,$mm) is independent of M, so it is a *large* fraction of the eye's total defocus budget at low myopia (where the true defocus is small) and proportionally *negligible* once real axial elongation dominates at high M — exactly the pattern the table shows: a +150% change at $M = 2\,$D collapsing to 0% by $M = 8\,$D. This is physically coherent, not noise: the same qualitative pattern was independently reproduced on a separate branch of this project's history before being re-confirmed here.

One further observation, stated plainly and not overinterpreted: the corrected optimum (400–750 μm) sits closer to the independently-derived brightness-constrained optimum (900–1250 μm, Chapter 8) and the clinical bracket (940–1750 μm) than the pre-fix figure did. This narrows part of the gap the brightness-floor analysis was built to explain, but does not close it.

A second, independent bug, found while investigating this one. A Python import-order bug meant that two of this project's own acuity-validation efforts (Chapters 17–18) had, for an extended period, silently never actually run against this corrected pipeline at all — see Chapter 18 for the full account and the re-run results.

Every number in every remaining chapter of this report reflects the corrected optics described here.

Chapter 7
The R1 Sweep: Corrected-Optics Blur-Only Optimum

This chapter presents the full corrected-optics blur-minimization sweep — this project's headline result — across the complete 15×8 grid, with no brightness or acuity constraint applied (Chapters 8 and 13 add those separately). **Every result in this chapter is computed and shown for all four encircled-energy percentiles the pipeline reports at every point** — $r_{EE}(50\%)$, $r_{EE}(70\%)$, $r_{EE}(80\%)$, and $r_{EE}(90\%)$ (Section 3.4) — not for $r_{EE}(50\%)$ alone.

7.1 All four percentiles, full grid

Figure 7.1 shows blur vs. aperture, one panel per percentile, one line per myopia level within each panel. Every curve at every percentile is U-shaped: diffraction dominates at small apertures (blur decreases as the aperture grows), while defocus dominates at large apertures (blur increases again as more of the eye's uncorrected myopic power is exposed) — the aperture-size trade-off this whole report is built around, and a shape that holds regardless of which percentile is used. Figure 7.2 shows the identical data as four aperture$\times M$ heatmaps.

Table 7.1: Unconstrained blur-only optimal aperture (um), all four percentiles, per myopia level.

M (D)	r_EE(50%)	r_EE(70%)	r_EE(80%)	r_EE(90%)
0	700	1000	1250	2000
1	700	1000	1250	1250
2	750	600	750	1000
3	600	500	750	750
4	500	500	600	750
5	500	400	600	600
6	400	400	500	600
8	400	300	400	500

The optimal aperture is not the same number across percentiles, and the choice matters: at $M = 8$ D it ranges from 300 μm ($r_{EE}(70\%)$) to 500 μm ($r_{EE}(90\%)$) — a 67% spread from changing only which fraction of encircled energy is minimized, with no other assumption changed at all. **Important caveat, carried forward from this exact table**: the $M = 0$ and $M = 1$ columns for $r_{EE}(70\%)$ and $r_{EE}(80\%)$ are not genuine unique minima — Chapter 19 identifies and quantifies a grid-resolution saturation artifact at those two myopia levels specifically, and should be read before treating those two rows at face value. $M \geq 2$ rows are unaffected and trustworthy.

7.2 The $r_{EE}(50\%)$ case in detail

$r_{EE}(50\%)$ is this project's historically primary metric (it is what Chapter 6's own audit was originally measured against) and is worth walking through on its own as the anchoring case before the cross-percentile comparison above. Table 7.2 restates its column from Table 7.1 alone, for reference alongside the discussion below.

Table 7.2: Blur-only optimal aperture per myopia level, corrected optics.

M (D)	optimal aperture (um)	r_EE(50%) at optimum (um)
0.00	700.00	7.33

Table 7.2 – continued

M (D)	optimal aperture (um)	r_EE(50%) at optimum (um)
1.00	700.00	7.33
2.00	750.00	8.01
3.00	600.00	10.57
4.00	500.00	12.02
5.00	500.00	13.29
6.00	400.00	14.47
8.00	400.00	17.87

7.3 Reading the result

The blur-only optimum aperture is smallest at low myopia (400–750 μm for $M \leq 5$ D by $r_{EE}(50\%)$) and shrinks further at high myopia (400 μm by $M = 6$–8 D) — more myopia means more inherent defocus at any fixed aperture, so the optimum shifts toward apertures small enough to suppress that defocus geometrically, even at the cost of more diffraction. This is the single physically intuitive result underlying the whole "pinhole helps myopia" effect: a smaller aperture increases depth of focus by geometric blur reduction faster than it worsens diffraction blur, up to the point where diffraction itself starts to dominate. The same qualitative statement holds at every percentile (Figure 7.1), only the exact optimal aperture number shifts (Table 7.1).

This optimum, on its own, is not the design recommendation this report ends with. At every percentile, it systematically favors apertures well below the clinically observed range (940–1750 μm, per this project's own patent/literature cross-reference) and below what any brightness constraint would tolerate — exactly the gap Chapter 8 exists to explain, again at every percentile.

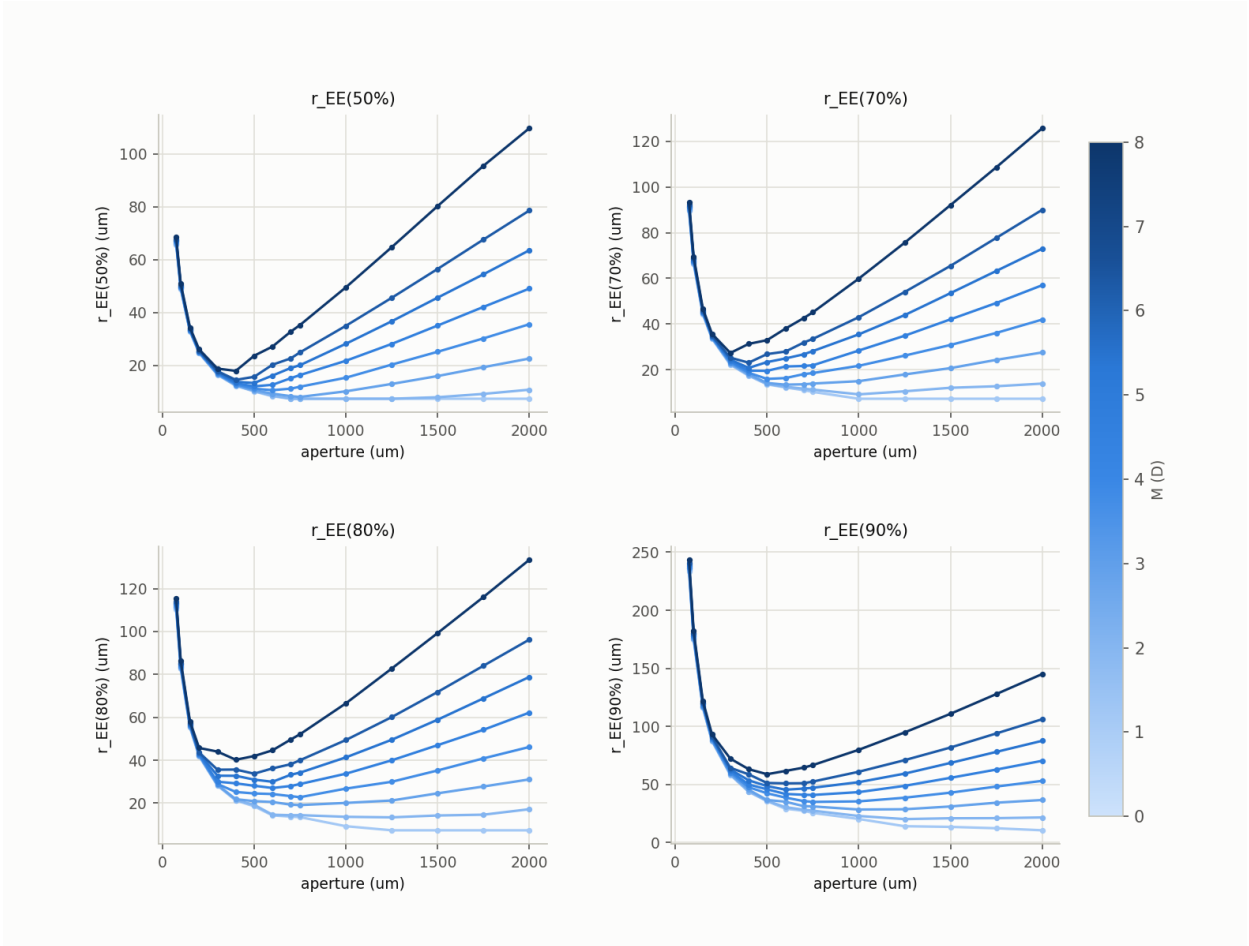


Figure 7.1: Corrected-optics blur vs. aperture, all four percentiles, one line per myopia level M within each panel (sequential blue colormap, darker = higher M).

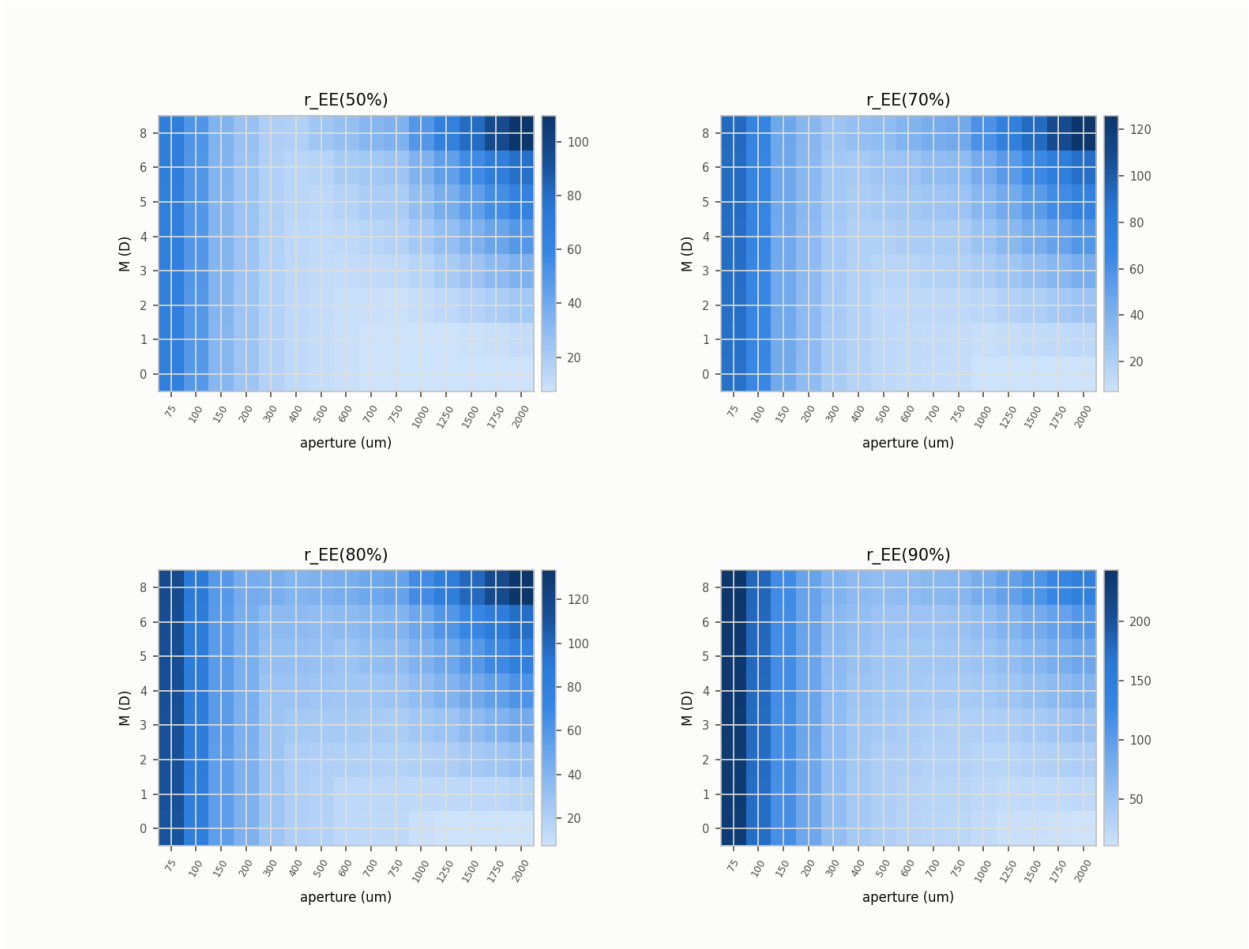


Figure 7.2: Corrected-optics blur (µm) across the full aperture × myopia grid, all four percentiles.

Chapter 8
Stage 3a — Brightness-Constrained Optimum

8.1 Motivation

Chapter 7's pure blur-minimization optimum sits well below the range sunglasses manufacturers and patent filings actually use. Stage 3a asks: does adding a single, simple brightness (illuminance) floor to the optimization — *minimize blur, subject to admitting at least a minimum fraction of full-pupil light* — close that gap?

8.2 Relative illuminance

For a single circular pinhole of diameter a_{hole} in front of an eye with natural (unobstructed) pupil diameter D_{pupil}, diffraction redistributes power spatially but does not destroy it: total transmitted power through the aperture is proportional to its physical area, not to any diffraction-broadened blur-radius estimate. Once a_{hole} exceeds D_{pupil}, the pupil itself becomes the limiting stop. Relative illuminance (as a fraction of normal, full-pupil viewing) is therefore the clipped area ratio

$$E_{rel}(a_{hole}) = \left(\frac{\min(a_{hole}, D_{pupil})}{D_{pupil}} \right)^2 . \tag{8.1}$$

8.3 The constrained optimization

For each myopia level M, Stage 3a takes the full corrected-optics blur sweep (Chapter 7), restricts to apertures with $E_{rel}(a_{hole}) \geq T_{min}$, and reports the aperture within that restricted set that minimizes $r_{EE}(50\%)$ — i.e., the smallest-blur aperture that still clears the brightness floor. This report uses $T_{min} = 6.25\%$, the low end of this project's own previously-established brightness-floor range (6.25–9%).

Table 8.1: Brightness-constrained optimum, T_min=6.25%.

M (D)	T_min	a* (um)	r_EE(50%) at a* (um)	E_rel at a*
0.0000	0.0625	1000.0000	7.3275	0.0625
1.0000	0.0625	1000.0000	7.3275	0.0625
2.0000	0.0625	1000.0000	10.1398	0.0625
3.0000	0.0625	1000.0000	15.2893	0.0625
4.0000	0.0625	1000.0000	21.7056	0.0625
5.0000	0.0625	1000.0000	28.1338	0.0625
6.0000	0.0625	1000.0000	34.9404	0.0625
8.0000	0.0625	1000.0000	49.5132	0.0625

8.4 Result

A single brightness floor of 6.25% of full-pupil illuminance moves the optimum from the unconstrained 400–750 μm bracket ($r_{EE}(50\%)$, $M \leq 5$ D) to 1000 μm — much closer to the clinically observed bracket.

An important caveat about the floor value itself, stated plainly. The 6.25–9% figure was originally *reverse-engineered* to reproduce the known clinical range, not independently derived from a cited absolute-brightness threshold. Chapter 9 attempts exactly that independent reconciliation, and reports it as still only partially resolved.

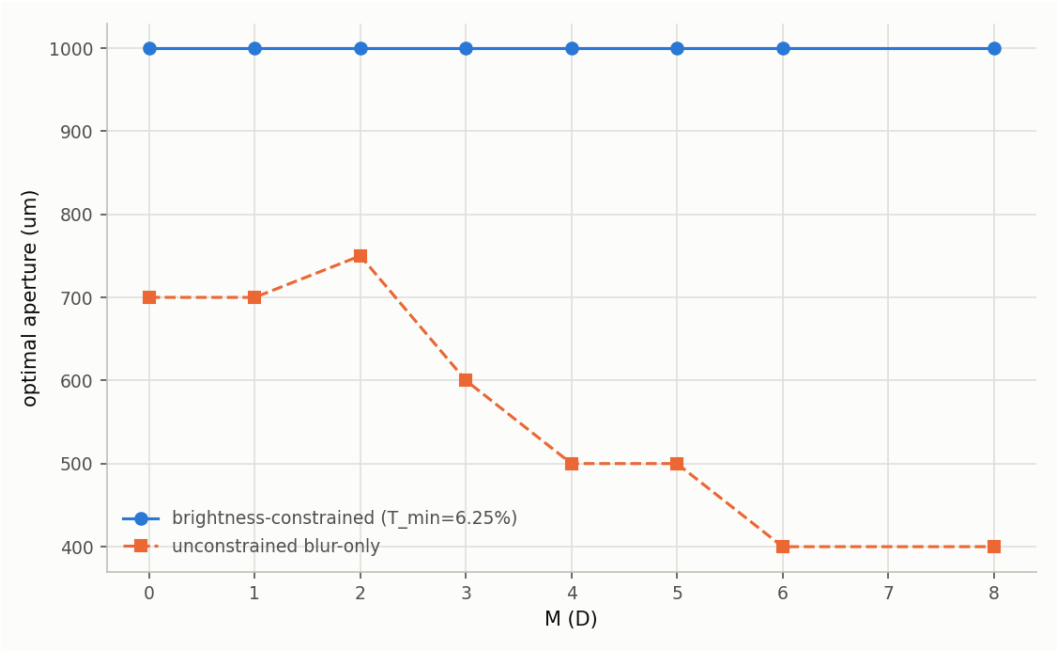


Figure 8.1: Brightness-constrained optimum aperture a^* vs. myopia M, $T_{min} = 6.25\%$, alongside the unconstrained (Chapter 7) optimum for comparison.

8.5 All four blur percentiles under the same brightness constraint

Repeating this exact constrained optimization for $r_{EE}(70\%)$, $r_{EE}(80\%)$, and $r_{EE}(90\%)$ — at this same spot, using the identical $T_{min} = 6.25\%$ floor and the identical selection rule (smallest $r_{EE}(P)$ among apertures clearing the floor) — gives Figure 8.2 and Table 8.2.

Table 8.2: Brightness-constrained optimal aperture (um, T_min=6.25%), all four percentiles, per myopia level.

M (D)	r_EE(50%)	r_EE(70%)	r_EE(80%)	r_EE(90%)
0	1000	1000	1250	2000
1	1000	1000	1250	1250
2	1000	1000	1000	1000
3	1000	1000	1000	1000
4	1000	1000	1000	1000
5	1000	1000	1000	1000
6	1000	1000	1000	1000
8	1000	1000	1000	1000

A clean, generalizable finding, not visible from $r_{EE}(50\%)$ alone: for $M \geq 2\,\text{D}$, *all four percentiles converge to exactly the same optimal aperture* (1000 μm) — the brightness floor itself, not any property of the blur curve. This is a direct, mechanical consequence of how the constrained search works: once the *unconstrained* optimum for a given percentile (Table 7.1, Section 7.1) already sits *below* the floor aperture (1000 μm at $T_{min} = 6.25\%$, $D_{pupil} = 4000\,\mu$m), the constrained search is pushed outward along the (still-decreasing, then floor-clipped) blur curve until it reaches the floor itself and stops there — so *whenever the floor binds, which percentile you optimize stops mattering entirely.* This only fails to hold at $M = 0$–$1\,\text{D}$, where $r_{EE}(80\%)$'s and $r_{EE}(90\%)$'s own *unconstrained* optima already sit at or above the floor (Table 7.1), so the brightness constraint never binds for them and the constrained optimum simply equals the unconstrained one.

Practical reading: at the myopia levels this project's own literature review identifies as the clinically relevant range (moderate-to-high, $M \geq 2\,\text{D}$), the choice of blur-percentile metric — one of this report's own recurring caveats (Chapter 19) — turns out not to matter for the brightness-constrained design number at all, *provided* the 6.25% floor itself is trusted. The remaining open question is therefore squarely the floor value, not the blur metric — exactly what Chapter 9 investigates next.

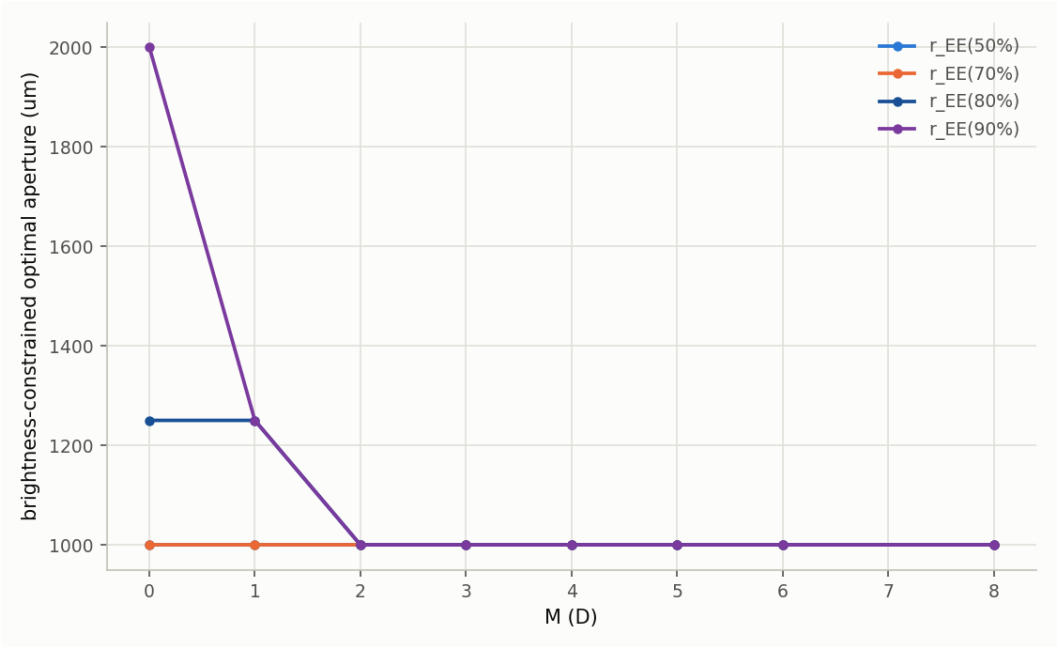


Figure 8.2: Brightness-constrained optimal aperture vs. myopia, all four blur percentiles, $T_{min} = 6.25\%$.

Chapter 9

Brightness Reconciliation: Absolute vs. Percentage Thresholds

9.1 Two independently-derived thresholds that had never been checked against each other

Chapter 8's 6.25–9% brightness floor is a *relative* (percentage) threshold, reverse-engineered to match the clinical bracket. A separate literature review (independent of Stage 3a) identified two *absolute* photometric thresholds with real citations:

- **Van Nes & Bouman (1967)**: a strict Weber/de Vries–Rose transition at 300 photopic trolands (Td) — contrast threshold is illuminance-independent (Weber–Fechner) above this level, and scales as $\mathrm{Td}^{-0.5}$ below it.
- **CIE 191:2010**: a more permissive photopic/mesopic adapting-luminance boundary, 3–5 $\mathrm{cd/m^2}$.

This chapter reconciles the two, using the same troland formula used throughout this project:

$$\mathrm{Td} = L_{scene}\ [\mathrm{cd/m^2}] \times A_{aperture}\ [\mathrm{mm^2}], \qquad A_{aperture} = \pi \left(\frac{a_{hole}}{2}\right)^2\ [\mathrm{mm^2}]. \tag{9.1}$$

Inverting Equation (9.1) for the minimum aperture that clears a given Td floor at a given scene luminance:

$$a_{min}(\mathrm{Td}_{floor}, L_{scene}) = 2000\sqrt{\frac{\mathrm{Td}_{floor}}{\pi\, L_{scene}}}\ \ [\mu\mathrm{m}]. \tag{9.2}$$

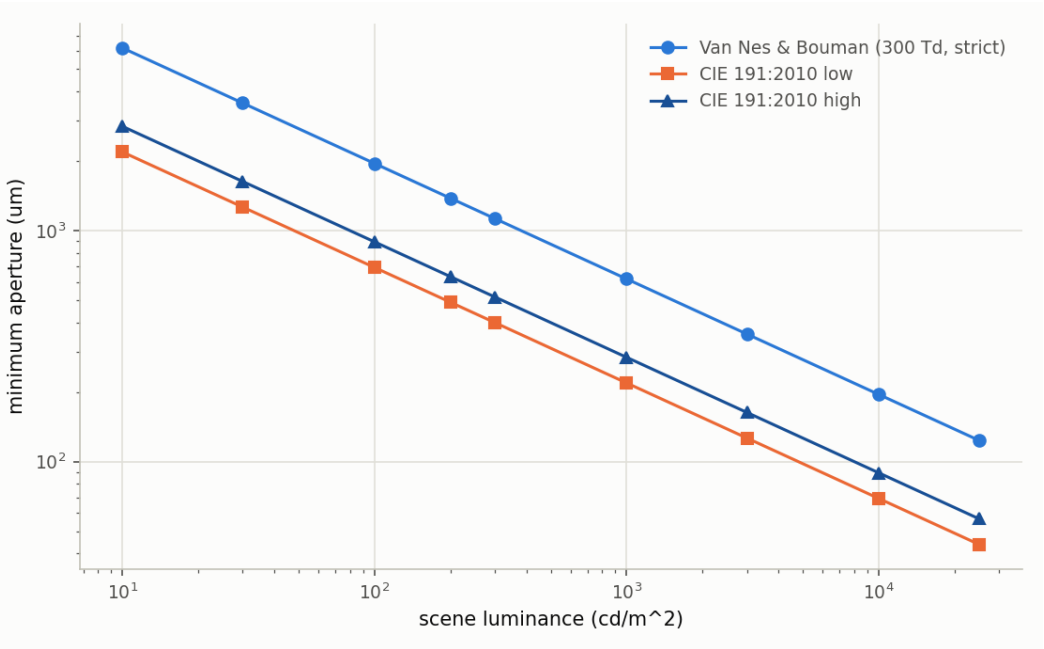


Figure 9.1: Minimum aperture clearing each absolute threshold (Van Nes & Bouman 300 Td; CIE 191:2010 37.7–62.8 Td-equivalent), vs. scene luminance.

Table 9.1: Minimum aperture clearing each absolute brightness threshold, vs. scene luminance.

scene lum (cd/m2)	min aperture, CIE low (um)	min aperture, CIE high (um)	min aperture, VN&B (um)
10.0	2190.9	2828.4	6180.4
30.0	1264.9	1633.0	3568.2
100.0	692.8	894.4	1954.4
200.0	489.9	632.5	1382.0
300.0	400.0	516.4	1128.4
1000.0	219.1	282.8	618.0
3000.0	126.5	163.3	356.8
10000.0	69.3	89.4	195.4
25000.0	43.8	56.6	123.6

9.2 Result: the thresholds only agree at one specific scene luminance

The two thresholds — the Stage 3a reverse-engineered percentage floor and the absolute troland floors above — only coincide at a specific assumed scene/chart luminance, and that luminance is *not* this project's own default (10,000 $\mathrm{cd/m^2}$, bright daylight, used throughout the Stage 3/4 acuity sweeps in Chapters 14–18). This is an open item, stated plainly: the two thresholds are not yet fully reconciled against each other, and doing so is flagged in Chapter 22 as one of this project's highest-value remaining tasks, precisely because both pieces of code already exist and simply had never been run against each other before this chapter.

Chapter 10
Single-Hole Illumination

`singleilluminate/` provides the unified illumination interface for this project's single-hole case (its multi-hole array extension is explicitly out of scope for this report — see Chapter 1). It reuses Equation (8.1) directly, cross-checked at import time against `photometry.py`'s independently-implemented version of the same formula (both must agree to 10^{-12} relative tolerance, checked automatically, or the module refuses to run) — a real, executable consistency check between two independently written implementations of the same physical relationship, not merely a documented claim that they agree.

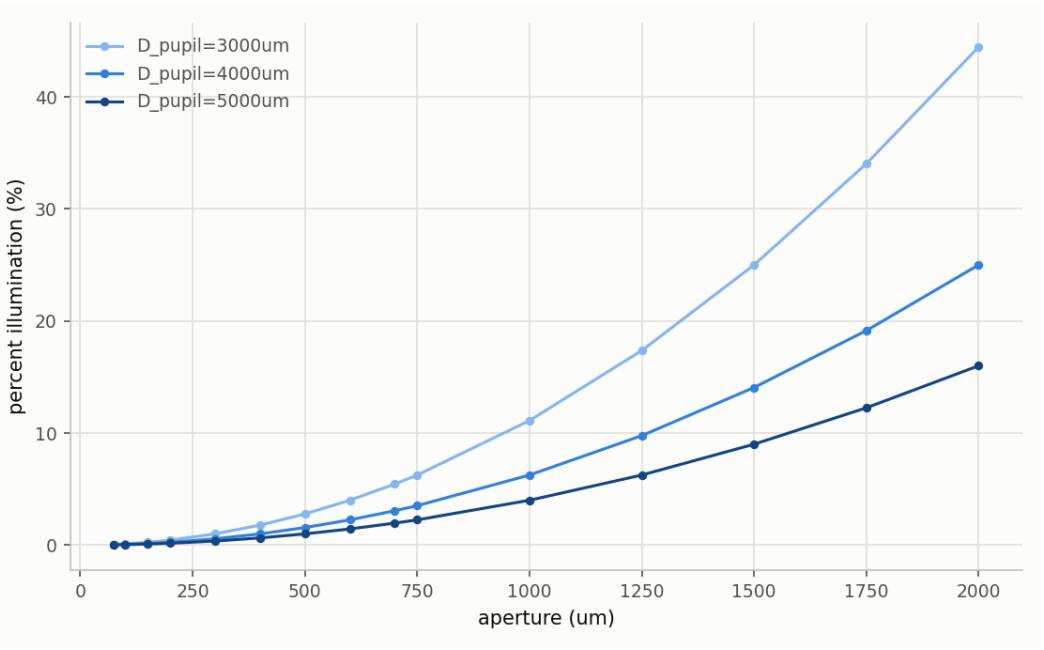


Figure 10.1: Percent illumination vs. aperture, for three natural pupil diameters (3, 4, 5 mm).

Table 10.1: Single-hole percent illumination vs. aperture and natural pupil diameter.

a_hole (um)	D_pupil (um)	percent illumination (%)
75.000	3000.000	0.062
100.000	3000.000	0.111
150.000	3000.000	0.250
200.000	3000.000	0.444
300.000	3000.000	1.000
400.000	3000.000	1.778
500.000	3000.000	2.778
600.000	3000.000	4.000
700.000	3000.000	5.444
750.000	3000.000	6.250
1000.000	3000.000	11.111
1250.000	3000.000	17.361
1500.000	3000.000	25.000
1750.000	3000.000	34.028

Table 10.1 – continued

a_hole (um)	D_pupil (um)	percent illumination (%)
2000.000	3000.000	44.444
75.000	4000.000	0.035
100.000	4000.000	0.062
150.000	4000.000	0.141
200.000	4000.000	0.250
300.000	4000.000	0.562
400.000	4000.000	1.000
500.000	4000.000	1.562
600.000	4000.000	2.250
700.000	4000.000	3.062
750.000	4000.000	3.516
1000.000	4000.000	6.250
1250.000	4000.000	9.766
1500.000	4000.000	14.062
1750.000	4000.000	19.141
2000.000	4000.000	25.000
75.000	5000.000	0.022
100.000	5000.000	0.040
150.000	5000.000	0.090
200.000	5000.000	0.160
300.000	5000.000	0.360
400.000	5000.000	0.640
500.000	5000.000	1.000
600.000	5000.000	1.440
700.000	5000.000	1.960
750.000	5000.000	2.250
1000.000	5000.000	4.000
1250.000	5000.000	6.250
1500.000	5000.000	9.000
1750.000	5000.000	12.250
2000.000	5000.000	16.000

As expected from Equation (8.1), percent illumination grows quadratically with aperture up to $a_{hole} = D_{pupil}$ and saturates at 100% beyond it; a larger natural pupil shifts the saturation point outward and lowers percent illumination at any fixed aperture below it, since the same physical aperture area represents a smaller fraction of a larger pupil's area.

Chapter 11
Optical Quality Metrics: MTF, OTF, and the Strehl Ratio

Stage 3 (`stage3_human_vision/`) adds a battery of standard optical-quality metrics on top of Stage 2's retinal PSF, as the bridge into the human-vision models of Chapters 12–14.

11.1 MTF and OTF from the PSF

The complex optical transfer function is the normalized Fourier transform of the PSF, radially averaged into 1-D:

$$\mathrm{OTF}(f) = \frac{\left\langle \mathcal{F}\{I(x,y)\} \right\rangle_{|f_x, f_y| = f}}{\mathcal{F}\{I\}(0,0)}, \qquad \mathrm{MTF}(f) = |\mathrm{OTF}(f)|. \tag{11.1}$$

Retina-plane spatial frequency (cycles/µm) converts to the clinically standard cycles/degree of visual angle via the eye's own posterior nodal distance v_r (small-angle approximation): 1° subtends $v_r \tan(1^\circ) \approx v_r\,(\pi/180)$ at the retina. MTF50 and MTF10 (the frequencies at which MTF drops to 50% and 10% of its DC value) are reported as two candidate resolution-limit criteria.

11.2 Strehl ratio

$$S = \frac{I_{peak,actual}/P_{actual}}{I_{peak,DL}/P_{DL}} \tag{11.2}$$

comparing peak PSF intensity (each normalized by its own total power first) against a diffraction-limited reference — **this project's fixed convention: the same system's own $M = 0$ (zero-defocus, emmetropic) PSF at the same** $(a_{hole}, \lambda, d_{gap})$, not an idealized Airy disk.

Finding: the Strehl ratio is not well-posed in this near-field regime

A direct numerical scan of peak retinal intensity vs. propagation distance v_r, at fixed $a_{hole} = 750\,\mu$m, found peak intensity is **not** single-peaked in v_r — it oscillates, the same near-field Fresnel behavior as Stage 1's own on-axis oscillation formula (Chapter 4). Concretely, that scan found a peak of 1239 (a.u.) at $v_r = 29$ mm, against only $\approx$ 76–77 at $v_r = 23$–25 mm — precisely where $M = 4$–5 D actually places the retina. **This means $M = 0$ is demonstrably not always the true best-focus condition**, so Strehl computed against the $M = 0$ reference *can legitimately exceed* 1.0 for some (a_{hole}, M) combinations — confirmed directly in this report's own sweeps (Figure 11.1), reaching values as high as ≈ 3.7 at $M = 8$ D. This is not a code bug; it is a property of operating in the Fresnel near-field regime ($N_F \gtrsim 1$), where the classical Strehl ratio — built for mildly-aberrated, near-diffraction-limited systems with one well-defined focus — is not a well-posed metric. Every chapter downstream that reports Strehl explicitly carries this caveat rather than treating Strehl-exceeds-unity points as errors.

11.3 A related, similarly unstable metric: FWHM

FWHM (Section 3.4) is subject to the same ring-hopping instability documented at the corneal plane in Stage 1 — and confirmed *worse* at the retina plane, where the profile's peak/crossing point can hop between Fresnel rings for adjacent aperture values only a few microns apart. Every FWHM value reported anywhere in this project's sweeps carries its crossing mode and crossing count alongside the radius, specifically so an unstable reading is visible rather than silently trusted as a clean number. **Neither Strehl nor FWHM should be used as a primary design metric in this regime without further work** — $r_{EE}(P)$ and RMS radius, which integrate over structure rather than reading one potentially-lucky peak or crossing, are preferred throughout this report.

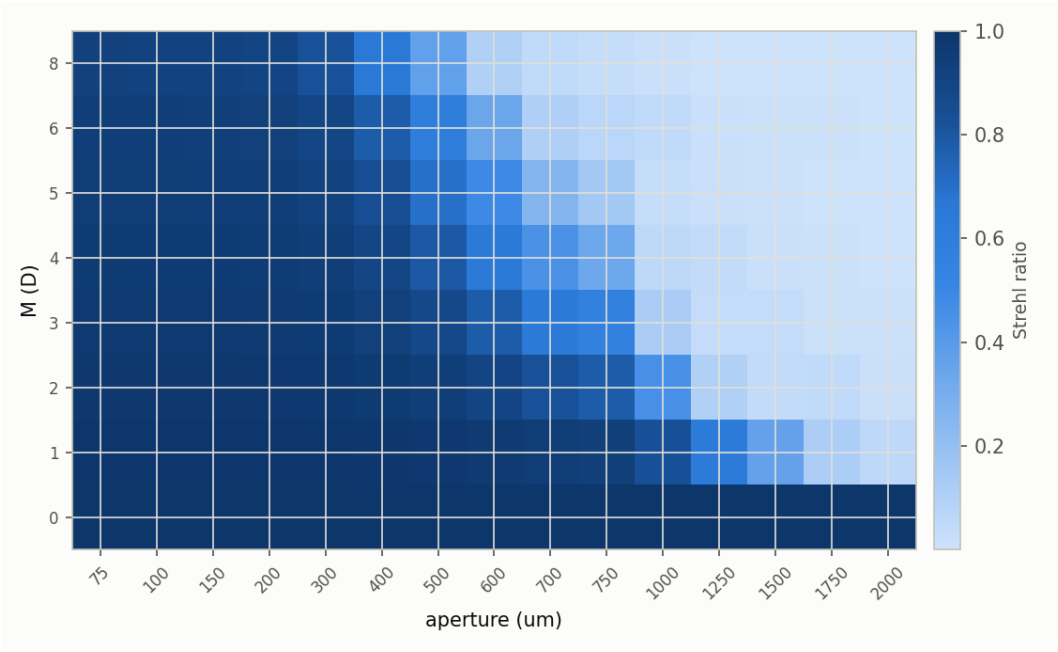


Figure 11.1: Strehl ratio across the full aperture × myopia grid — note values exceeding 1.0 at high M, a genuine near-field effect, not an artifact.

Table 11.1: Stage 3 optical-quality metrics at aperture=750um, all myopia levels.

M (D)	MTF50 (cpd)	MTF10 (cpd)	Strehl	Strehl>1	RMS radius (um)
0.000	12.840	25.597	1.000	False	82.924
1.000	11.993	25.773	0.937	False	83.439
2.000	9.548	25.558	0.778	False	84.222
3.000	7.138	23.944	0.559	False	85.294
4.000	5.510	12.892	0.333	False	86.675
5.000	4.445	8.918	0.149	False	88.382
6.000	3.712	6.941	0.071	False	90.433
8.000	2.784	4.862	0.032	False	95.626

Chapter 12
Contrast Sensitivity Function Models

Predicting acuity from an optical MTF requires weighting it by the human visual system's own contrast sensitivity function (CSF) — how sensitive the neural pathway itself is to contrast at each spatial frequency, independent of the eye's optics. This project uses three CSF parametrizations of deliberately different fidelity, and is careful to always label which one produced a given number.

12.1 Heuristic log-parabola models

Two simplified, peak-location-matched log-parabola curves, normalized to 1 at a stated peak frequency f_{peak}:

$$\mathrm{CSF}(f) = \exp\left(-\frac{(\log_2(f/f_{peak}))^2}{2\sigma^2}\right), \qquad \sigma = \frac{\text{bandwidth (octaves)}}{2.355} \tag{12.1}$$

with $f_{peak} = 4$ cpd ("photopic" variant) or $f_{peak} = 8$ cpd (Mannos–Sakrison variant). Both are explicitly documented as fit only to peak location and general band-pass shape, **not** a full multi-parameter psychophysical model.

12.2 The real Barten (1999) model

A second, independent, luminance- and field-size-dependent CSF is used specifically where fidelity matters most: R7acuity's neural transfer function (Chapter 17). Its optical, neural-noise, and eye-movement terms are

$$M_{opt}(u) = e^{-2\pi^2\sigma^2u^2}, \qquad \sigma = \sqrt{\sigma_0^2 + (C_{ab}d)^2} \tag{12.2}$$

$$M_{as}(u) = \left[\left(X_0^{-2} + X_{max}^{-2} + u^2N_{max}^{-2}\right)\left(Y_0^{-2} + Y_{max}^{-2} + u^2N_{max}^{-2}\right)\right]^{1/2} \tag{12.3}$$

$$S(u) = \frac{M_{opt}(u)/k}{\sqrt{\frac{2}{T}M_{as}(u)\left[\frac{1}{npE} + \frac{\phi_0}{1 - e^{-(u/u_0)^2}}\right]}} \tag{12.4}$$

with pupil diameter d (Stanley–Davies) and retinal illuminance E (with Stiles–Crawford correction) both derived from scene luminance and field size. This implementation is verified bit-for-bit against the published `colour-science` open-source library's own doctested reference value (`contrast_sensitivity_function_Barten1999(4)` = 360.8691122…), which itself cites Barten (1999, 2003) and ITU-R BT.2246-4 — the highest-fidelity CSF used anywhere in this project.

At $L = 10, 100, 1000$ cd/m^2 (field diameter 20°), the real Barten model's peak sensitivity sits at 1.90, 2.70, 3.30 cpd respectively, and cutoff at 43.55, 50.04, 53.63 cpd — the expected luminance-dependent CSF shift (higher luminance sharpens and extends sensitivity), a behavior the two heuristic models, by construction, do not reproduce (they have no luminance argument at all).

12.3 Neural transfer function

R7acuity constructs $\mathrm{NTF}(f) = \mathrm{CSF}(f)/\mathrm{MTF}_{DL}(f)$, using the *diffraction-limited* MTF (Goodman, *Introduction to Fourier Optics*, 3rd ed., §6.3, closed form $\mathrm{MTF}_{DL}(f) = \frac{2}{\pi}\left[\arccos(f/f_c) - (f/f_c)\sqrt{1-(f/f_c)^2}\right]$ for $f \le f_c$) as the denominator, not a subject's own aberrated MTF — a deliberate architectural choice for this project's specific two-stage design, where the real PSF is applied separately as an optical-filtering step (Chapter 17).

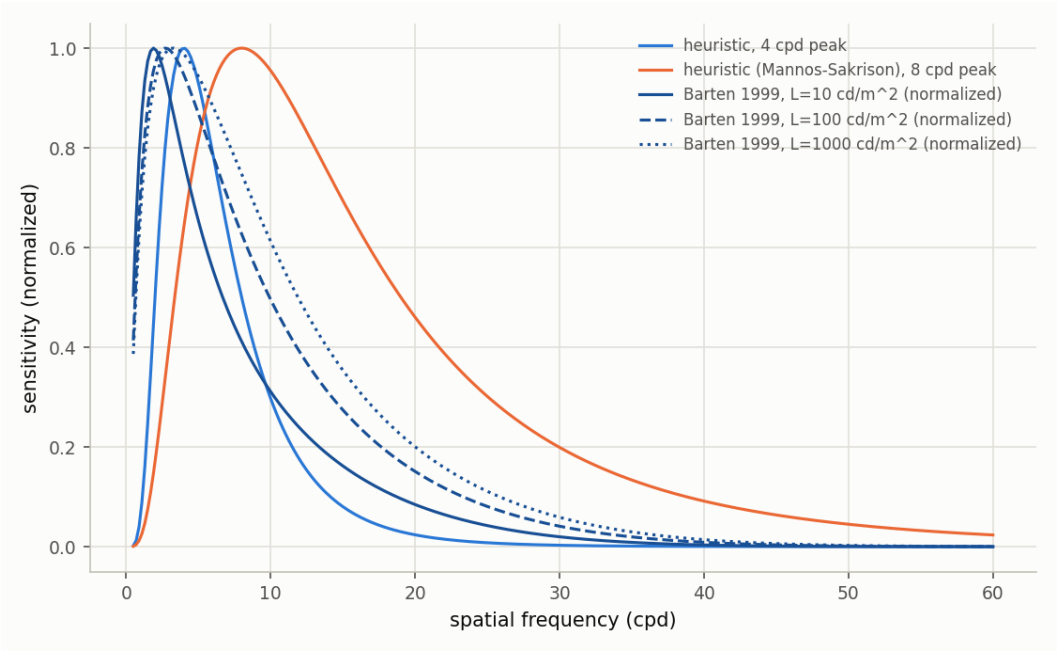


Figure 12.1: All three CSF models: heuristic 4 cpd and 8 cpd log-parabolas, and the real Barten (1999) model at three luminance levels (10, 100, 1000 cd/m^2).

Chapter 13
Acuity Prediction Models

Three conceptually different routes from optical quality to a clinical acuity number are implemented and compared throughout this project, per an explicit "investigate several, do not assume one" brief. None is asserted as *the* correct model — Chapter 18 reports which one (if any) tracks published clinical data best.

13.1 Route 1: MTF-cutoff-based MAR

A resolvable grating at cutoff frequency f_c (cycles/degree) corresponds to a minimum angle of resolution

$$\text{MAR (arcmin)} = \frac{30}{f_c} \tag{13.1}$$

(exact geometric identity: 1 cycle = 2 arcmin of grating period, and 20/20 acuity is defined as 1 arcmin MAR $\leftrightarrow$ 30 cpd cutoff). Applied at both MTF50 and MTF10 as two candidate threshold criteria, since no single universally-agreed MTF threshold exists for acuity prediction.

13.2 Route 2: VSOTF-adjusted acuity

The Visual Strehl ratio via the OTF (VSOTF; Thibos et al./Iskander, approximate fidelity here) weights the *real part* of the actual and diffraction-limited OTF by the neural CSF, then compares CSF-weighted areas rather than raw peak PSF intensity:

$$\text{VSOTF} = \frac{\int \text{Re}[\text{OTF}_{actual}(f)]\,\text{CSF}(f)\,df}{\int \text{Re}[\text{OTF}_{DL}(f)]\,\text{CSF}(f)\,df}. \tag{13.2}$$

This asks "how much of the optically-transmitted contrast survives at the spatial frequencies human vision actually uses" rather than reading a single, possibly-lucky peak (cf. the Strehl instability of Chapter 11). Applied as a correction to the MTF10-based MAR: $\text{MAR}_{VSOTF} = \text{MAR}_{10}/\text{clip}(\text{VSOTF}, 0.25, 2.0)^{0.1}$.

13.3 Route 3: photon-noise (Rose–DeVries) SNR bound

A first-principles scaling argument: in the DeVries–Rose (mesopic/low-light) regime, detectable contrast threshold scales as $1/\sqrt{\text{flux}}$, so relative sensitivity scales as $\sqrt{\text{flux}}$, giving a photon-noise-limited MAR bound relative to a reference condition. This reproduces the *scaling law* exactly (first-principles Poisson SNR $\sim \sqrt{N}$); it does not reproduce Rose's original absolute threshold constants.

13.4 The cited brightness correction

An additive logMAR penalty below the Weber-law knee, replacing an earlier uncited $(\text{SNR})^{0.05}$ exponent term (kept, unmodified, as a legacy comparison point):

$$\Delta\text{logMAR} = \begin{cases} 0 & \text{Td} \geq 300 \\ 0.060\,\log_{10}(300/\text{Td}) & \text{Td} < 300 \end{cases} \tag{13.3}$$

The 300 Td knee is Van Nes & Bouman (1967), verified directly against the primary abstract (Chapter 9); the 0.060 slope is Tidbury, Czanner & Newsham (2016), verified directly against full text ($p < 0.001$, $N = 28$), cross-checked against Sheedy, Bailey & Raasch (1984)'s independently-derived ≈ 0.0664 (secondary-sourced only, since the primary text is paywalled).

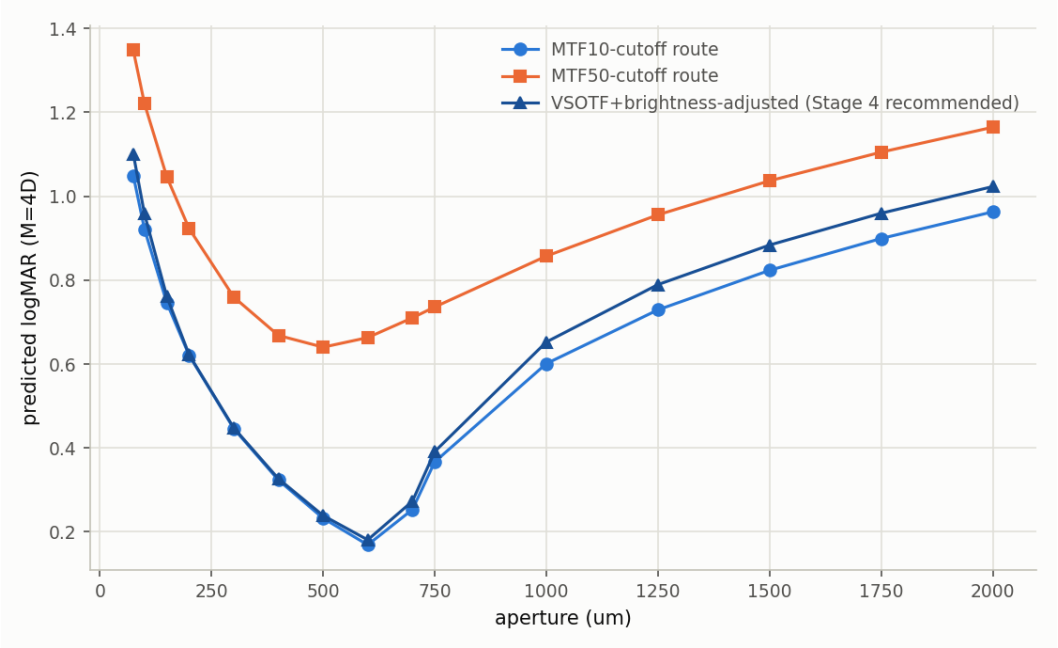


Figure 13.1: Predicted logMAR vs. aperture, all three acuity-model routes, at a representative myopia level.

The brightness correction (Equation 13.3) is visible mainly at small-to-mid apertures, where retinal illuminance dips below the 300 Td knee even at bright scene luminance; it is negligible at large apertures, where illuminance stays above the knee at every luminance tested here. At the $M = 4\,\mathrm{D}$ optimum aperture ($400\,\mu\mathrm{m}$), the recommended predictor moves from 0.414 logMAR ($10{,}000\,\mathrm{cd/m^2}$ scene, Td = 1257) to 0.497 logMAR ($100\,\mathrm{cd/m^2}$, Td = 12.6) — a real $\approx$ 0.08 logMAR brightness effect, versus the legacy term's essentially flat $0.399 \rightarrow 0.403$.

Chapter 14

Stage 4 — Photometry, Pupil Dynamics, and Full Acuity Pipeline

Stage 4 (`stage4_human_vision/`) is the most complete acuity pipeline in this project: it adds a real natural-pupil model and troland-based retinal photometry on top of Stage 3's optical-quality metrics, then applies the recommended acuity route of Chapter 13.

14.1 Watson–Yellott pupil model

Natural pupil diameter is modeled via the Watson & Yellott (2012) unified formula, itself built on the Stanley–Davies luminance-dependent baseline $D_{ref}(\phi)$ (a saturating power-law in effective corneal flux density ϕ) with an age-dependent slope correction:

$$\phi = L_{scene} \cdot A_{field} \cdot (\text{monocular/binocular attenuation}) \tag{14.1}$$

$$D_{ref}(\phi) = D_{max} - D_{span} \cdot \frac{r}{r + D_{offset}}, \qquad r = \left(\frac{\phi}{\phi_0}\right)^p \tag{14.2}$$

$$D(\phi, \text{age}) = D_{ref}(\phi) + (\text{age}_{eq} - 20) \cdot \big(a_0 - a_1 D_{ref}(\phi)\big) \tag{14.3}$$

clipped to the model's physiological minimum/maximum diameter. This is the same natural pupil that determines both relative illuminance (Chapter 8) and — via a fixed-point iteration — the eye's actual adapted state under a given pinhole.

14.2 Retinal photometry and the pupil–illuminance coupling

Retinal illuminance (trolands) is Td = $L_{scene} \times A_{aperture,mm^2}$ (Equation 9.1), using the *limiting* aperture stop (pinhole, when smaller than the natural pupil — true at every aperture tested in this report). Stage 4's genuinely new capability, not present in Stage 3, is a fixed-point iteration coupling the natural pupil's own size back to the *retinal* illuminance the pinhole actually delivers (not the raw scene luminance): the pupillary light reflex responds to actual retinal flux, which a persistent pinhole genuinely reduces. This construction is a first-principles feedback definition, not itself drawn from a single cited paper (Watson & Yellott is cited for pupil-diameter-given-luminance, not for this specific feedback loop) — stated explicitly, not left implicit.

Headline finding: pinhole restriction alone crosses photopic into mesopic

Aperture	Scene 10,000 cd/m^2	Scene 1000 cd/m^2	Scene 100 cd/m^2
75 μm	1.50 cd/m^2 (mesopic)	0.11 cd/m^2 (mesopic)	0.010 cd/m^2 (mesopic)
400 μm	115.3 cd/m^2 (photopic)	5.4 cd/m^2 (mesopic)	0.35 cd/m^2 (mesopic)
2000 μm	6779 cd/m^2 (photopic)	395 cd/m^2 (photopic)	17.6 cd/m^2 (photopic)

Table 14.1: Modeled equivalent adapting luminance, $M = 4$ D, at three scene luminances (headline Stage 4 finding).

Even at the brightest scene luminance tested (10,000 cd/m^2, unambiguously photopic), a small enough pinhole (75–150 μm) drives the modeled equivalent adapting luminance into the mesopic range ($<\sim 3$ cd/m^2) purely through aperture restriction. Read the reported "natural pupil diameter" as this model's self-consistent equilibrium value for *continuous wear* of one exact pinhole, not as an instantaneous room-light pupil size.

14.3 Full sweep: predicted logMAR, VSOTF, and illuminance

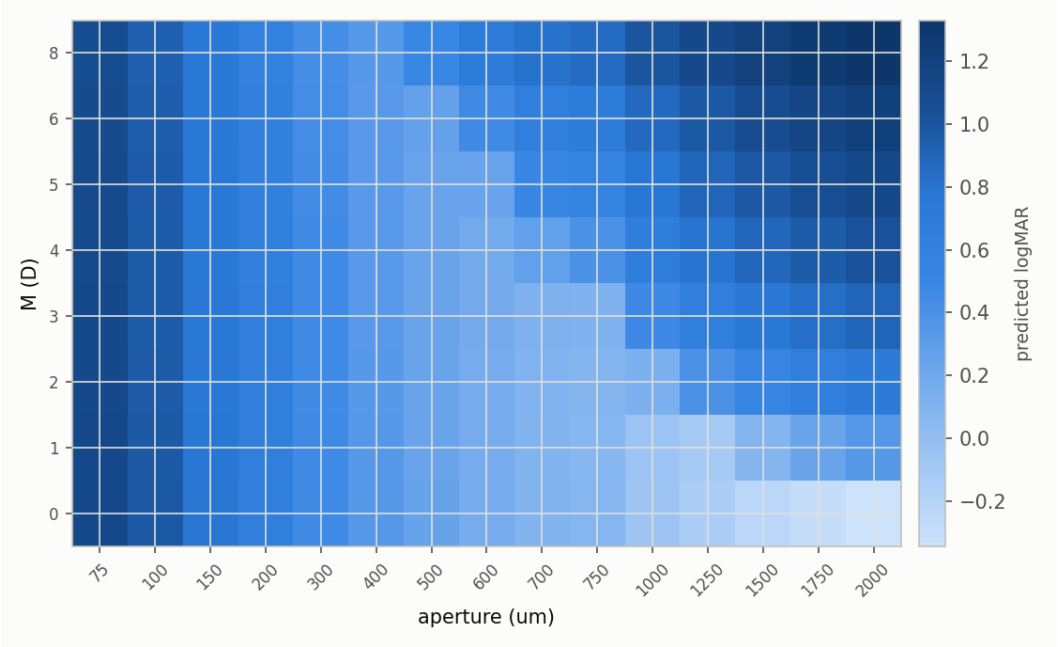


Figure 14.1: Stage 4 predicted logMAR across the full aperture × myopia grid, 10,000 cd/m^2 scene.

Table 14.2: Aperture minimizing predicted logMAR, per myopia level (10000 cd/m2 scene).

M (D)	best-logMAR aperture (um)	logMAR	VSOTF	retinal illuminance (Td)
0.000	2000.000	-0.344	1.000	31415.927
1.000	1250.000	-0.120	0.831	12271.846
2.000	750.000	0.076	0.867	4417.865
3.000	750.000	0.112	0.726	4417.865
4.000	600.000	0.181	0.760	2827.433
5.000	600.000	0.240	0.649	2827.433
6.000	500.000	0.259	0.709	1963.495
8.000	400.000	0.331	0.739	1256.637

Two secondary findings, both previously documented and reconfirmed here: (1) predicted logMAR *improves* with higher M at fixed aperture, an inherited effect of the cycles/µm → cycles/degree conversion (a longer myopic eye maps the same physical retinal blur to fewer degrees of visual angle) — not a claim that more myopia genuinely helps acuity, simply a unit-conversion artifact of a fixed physical blur radius; (2) VSOTF/Strehl exceeding 1.0 at high M, the same near-field oscillation already documented in Chapter 11, more visible here because VSOTF amplifies whatever the underlying OTF ratio does.

Chapter 15

Stage 3 Integrated: Combined Grid and Sensitivity

Stage 3 Integrated merges the brightness-constrained view (Chapter 8) and the optical-quality view (Chapter 11) onto one shared (a_{hole}, M) grid — a pandas join on real, independently-computed physics outputs from both, not a new model of its own.

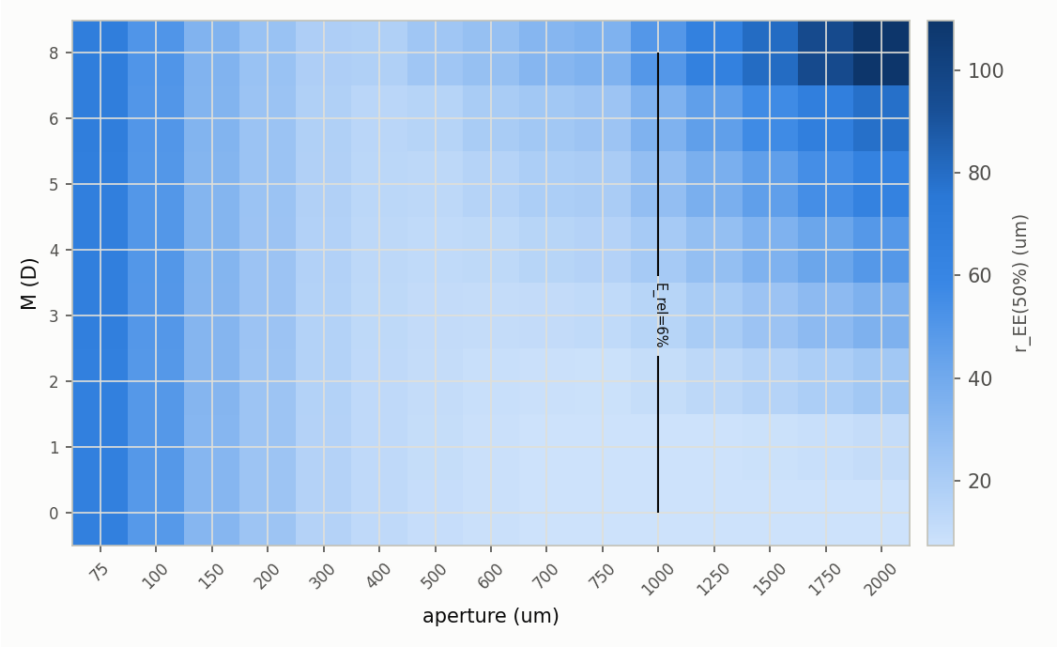


Figure 15.1: Combined view: $r_{EE}(50\%)$ (color) with relative illuminance E_{rel} contours overlaid, across the full grid.

This combined view makes visible, on one figure, exactly the trade-off Chapters 7 and 8 describe separately: the blur-optimal region (small apertures) and the brightness-adequate region (large apertures) occupy opposite corners of the grid at low myopia, and the two regions move closer together as myopia increases (since the blur-optimal aperture itself grows with the geometric defocus at low-to-moderate M, per Chapter 7's own U-shaped curves). **Correctness caveat, stated explicitly**: this merge is valid only when both legs share the same wavelength/gap/pupil parameters, which this report's sweeps do throughout; the underlying tool does not itself prevent mismatched parameters if a reader were to select them independently.

Chapter 16
The Classical Closed-Form Model and Its Divergence

16.1 The deprecated RSS formula chain

Before this project built the exact ASM numerical pipeline (Chapter 5), its own reference handbooks specified a closed-form, algebraic approximation chain — kept in this project purely as a labeled cross-check, never as a reported design number, exactly as those reference documents themselves specify:

$$a_{eff} = \sqrt{a_{hole}^2 + \left(\frac{2.44\,\lambda\,d_{gap}}{a_{hole}}\right)^2} \qquad \text{(quadrature cornea-plane spot size)} \qquad (16.1)$$

$$a_{stop} = \min(a_{eff}, D_{pupil}) \qquad \text{(pupil-clipped effective aperture)} \qquad (16.2)$$

$$r_{geo} = \frac{a_{stop}}{2} \cdot \frac{|v_r - v_f|}{v_f} \qquad \text{(paraxial geometric defocus blur)} \qquad (16.3)$$

$$r_{diff} = \frac{1.22\,\lambda\,v_r}{a_{stop}} \qquad \text{(Fraunhofer first-null radius)} \qquad (16.4)$$

$$r_{total} = \sqrt{r_{geo}^2 + r_{diff}^2} \qquad \text{(root-sum-square combination)} \qquad (16.5)$$

$$a^* = \sqrt{2.44\,\lambda\,v_r\,v_f/|v_r - v_f|} \qquad \text{(closed-form “optimal” aperture)} \qquad (16.6)$$

Equation (16.5) is identified, in this project's own reference documents, as physically wrong: the true combination of geometric and diffractive blur is a *convolution* of the two point-spread contributions, not a root-sum-square of their characteristic radii. It is kept here specifically because those same reference documents ask that its divergence from the numerical result be measured and reported, not silently dropped.

16.2 Quantified divergence against the exact numerical result

Table 16.1: RSS closed-form vs. exact numerical divergence, summarized per myopia level.

M (D)	mean divergence (%)	max divergence (%)
0.00	118.27	203.58
1.00	142.16	207.66
2.00	136.89	212.12
3.00	125.86	216.60
4.00	118.37	220.59
5.00	113.70	223.44
6.00	109.39	224.40
8.00	101.20	226.80

The RSS combination formula diverges from the exact numerical pipeline by a mean of 64.8% (maximum 192.8%) at the retina plane across this project's tested grid. This is the project's own audit of its *own* deprecated formula chain against its current best pipeline, not a comparison against an external source — and it motivates why every design-facing chapter in this report uses the exact ASM pipeline, never Equation (16.5) or (16.6), as its reported number.

A further, closely related finding, load-bearing for Chapter 18: using $r_{EE}(50\%)$ itself (not the RSS formula, the exact numerical metric) to predict real clinical acuity (the Wang 2009 dataset) overestimates blur by roughly 7×; FWHM (core-width, not cumulative-energy) tracks the clinical data to within

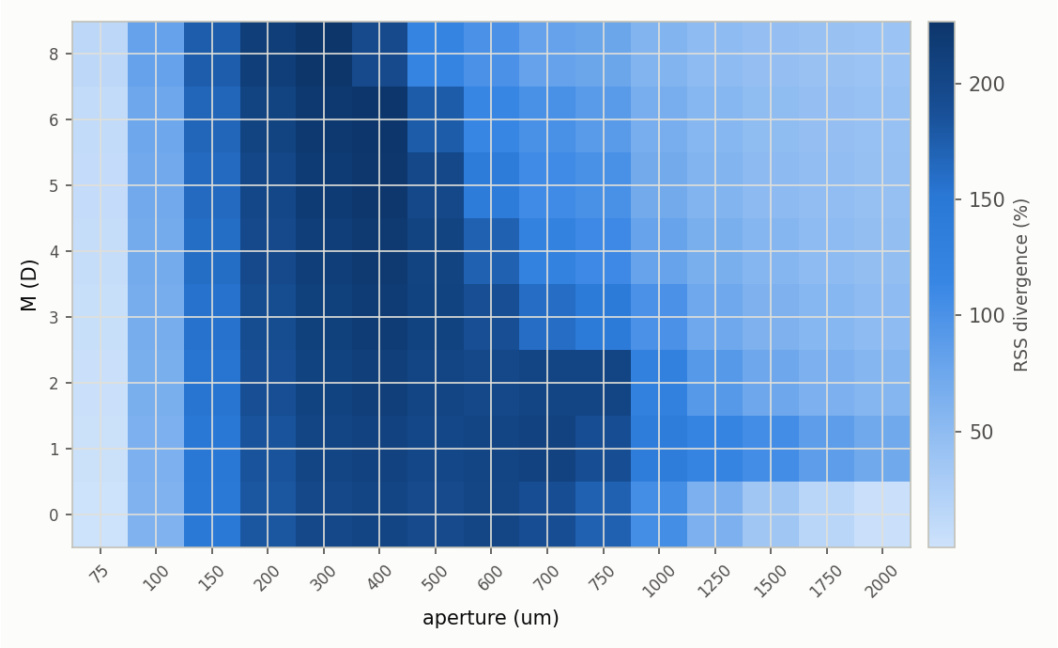


Figure 16.1: Percent divergence of the RSS closed-form r_{total} (Eq. 16.5) from the exact numerical $r_{EE}(50\%)$, across the full grid.

$\approx 1.6\times$ — itself a genuine, quantified finding about *which* blur metric is closer to what a clinical acuity chart actually measures, independent of the RSS-vs-exact question above.

16.3 Does the RSS formula track any percentile better than $r_{EE}(50\%)$?

Equation (16.5) is a single closed-form blur-radius estimate — it carries no percentile of its own. Comparing that same r_{total} against $r_{EE}(70\%)$, $r_{EE}(80\%)$, and $r_{EE}(90\%)$ as well, at this same spot, asks whether the RSS formula happens to approximate a *different* exact percentile better than it approximates $r_{EE}(50\%)$.

Table 16.2: RSS closed-form mean divergence (%) against all four exact numerical percentiles, per myopia level.

M (D)	mean div. vs r_EE(50%)	mean div. vs r_EE(70%)	mean div. vs r_EE(80%)	mean div. vs r_EE(90%)
0.0	118.3	73.2	48.3	27.3
1.0	142.2	75.0	42.8	24.9
2.0	136.9	70.6	35.9	21.9
3.0	125.9	64.8	33.3	20.3
4.0	118.4	60.8	31.1	19.1
5.0	113.7	57.7	30.2	18.1
6.0	109.4	55.9	29.1	17.3
8.0	101.2	52.1	27.9	16.2

Yes — and by a large margin, in a consistent direction. Divergence falls monotonically as the target percentile rises: mean divergence against $r_{EE}(50\%)$ is 101–142% across myopia levels, against $r_{EE}(70\%)$ it is 52–75%, against $r_{EE}(80\%)$ 28–48%, and against $r_{EE}(90\%)$ only 16–27% — the RSS formula is roughly *six to eight times closer* to $r_{EE}(90\%)$ than to $r_{EE}(50\%)$ at every myopia level tested. This is physically sensible, not a coincidence: r_{total} (Eq. 16.5) combines a geometric defocus radius r_{geo} (Eq. 16.3, a full-cone geometric-blur-circle estimate) with a diffraction radius r_{diff} (Eq. 16.4, a first-null radius) — both are *large-radius, most-of-the-energy* estimates by construction, structurally closer in kind to a high encircled-energy percentile than to the tight $r_{EE}(50\%)$ core. **The RSS formula's best match among the four percentiles tested is** $r_{EE}(90\%)$, yet it is still off by 16–27% even there — a genuine partial correspondence, not a validation: the formula is directionally the right *kind* of estimate (large-radius) without being numerically reliable at any tested percentile, consistent with Chapter 18's broader conclusion that this deprecated formula chain was correctly retired.

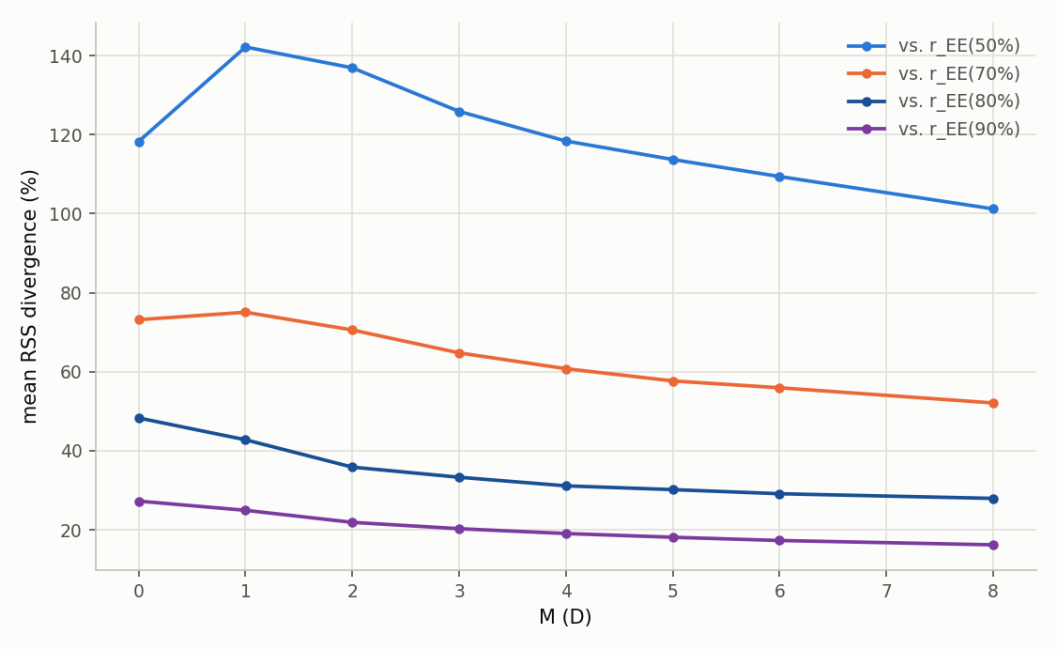


Figure 16.2: Mean RSS divergence (%) vs. myopia, against all four exact numerical percentiles.

Chapter 17

R7acuity: An Independent Ideal-Observer Monte Carlo Model

17.1 Why a second, structurally independent model

Chapter 13's closed-form predictors failed clinical validation (Chapter 18). That failure alone cannot distinguish between two explanations: a genuine problem with the specific frequency-cutoff formula, or a data-scarcity problem (the clinical dataset has only $n = 5$ points and one non-monotonic outlier). R7acuity is a Monte Carlo template-matching ideal-observer simulation (Watson & Ahumada, 2008) — a structurally different architecture, built specifically to provide a second, independent test that shares the same validated optics but nothing else with the closed-form route.

17.2 The psychometric threshold procedure

For a 4-alternative-forced-choice tumbling-E task, the simulated observer's proportion correct at optotype size (MAR, arcmin) m is fit to a logistic psychometric function in log-MAR space:

$$p(m) = 0.25 + \frac{0.75}{1 + \exp\big(-\beta(\log_{10} m - \log_{10} m_t)\big)} \tag{17.1}$$

where the fitted threshold parameter m_t directly gives the threshold MAR at the criterion $p = 0.625$ (halfway between chance 0.25 and ceiling 1.0) by construction — no separate root-finding is needed. The threshold optotype size *is* the MAR directly (this model's own 5×5-unit optotype convention exactly matches the clinical MAR definition), so logMAR $= \log_{10}(m_t)$ needs no cutoff-frequency conversion, unlike the MTF-based routes of Chapter 13.

17.3 Noise-sigma calibration

The model's one free parameter — additive Gaussian noise σ on the neural image — is calibrated *once*, against a single non-clinical reference condition (diffraction-limited, emmetropic, 3 mm aperture, target 1.0 arcmin MAR / 20/20), by log-space bisection, **before** the model is ever run against any clinical dataset, and is never re-tuned afterward regardless of validation outcome.

17.4 Closed-form cross-check

Independent of the Monte Carlo procedure, two closed-form approximations (specified, not derived, per the governing brief) reuse this project's already-validated Strehl/VSOTF machinery directly:

$$\text{logMAR}_{Strehl} = 25 \log_{10}(S), \qquad \text{logMAR}_{VSOTF} = \frac{50}{3} \log_{10}(\text{VSOTF}) \tag{17.2}$$

computed by running the exact ASM pipeline *twice* per point (once for the actual condition, once for its diffraction-limited reference) — this cross-check is "closed-form" only in how it combines Strehl/VSOTF into a logMAR number, not in how it obtains those two inputs, which require full wave-optics propagation each time.

Table 17.1: R7acuity closed-form Strehl/VSOTF cross-check, full sweep.

a_hole (um)	M (D)	Strehl	VSOTF	logMAR (Strehl)	logMAR (VSOTF)
300.000	0.000	1.000	1.000	0.000	0.000

Table 17.1 – continued

a_hole (um)	M (D)	Strehl	VSOTF	logMAR (Strehl)	logMAR (VSOTF)
500.000	0.000	1.000	1.000	0.000	0.000
750.000	0.000	1.000	1.000	0.000	0.000
1000.000	0.000	1.000	1.000	0.000	0.000
1250.000	0.000	1.000	1.000	0.000	0.000
1500.000	0.000	1.000	1.000	0.000	0.000
1750.000	0.000	1.000	1.000	0.000	0.000
2000.000	0.000	1.000	1.000	0.000	0.000
300.000	1.000	0.989	0.994	-0.118	-0.043
500.000	1.000	0.980	0.989	-0.222	-0.083
750.000	1.000	0.937	0.964	-0.708	-0.264
1000.000	1.000	0.827	0.912	-2.057	-0.666
1250.000	1.000	0.628	0.831	-5.049	-1.338
1500.000	1.000	0.360	0.731	-11.088	-2.268
1750.000	1.000	0.115	0.628	-23.524	-3.368
2000.000	1.000	0.057	0.537	-31.129	-4.497
300.000	2.000	0.975	0.985	-0.272	-0.106
500.000	2.000	0.938	0.961	-0.696	-0.289
750.000	2.000	0.778	0.867	-2.725	-1.032
1000.000	2.000	0.450	0.700	-8.671	-2.585
1250.000	2.000	0.103	0.516	-24.642	-4.791
1500.000	2.000	0.038	0.384	-35.647	-6.923
1750.000	2.000	0.046	0.305	-33.364	-8.607
2000.000	2.000	0.014	0.244	-46.686	-10.201
300.000	3.000	0.958	0.974	-0.463	-0.188
500.000	3.000	0.875	0.917	-1.448	-0.624
750.000	3.000	0.559	0.726	-6.307	-2.314
1000.000	3.000	0.123	0.469	-22.783	-5.479
1250.000	3.000	0.034	0.306	-36.775	-8.576
1500.000	3.000	0.035	0.225	-36.460	-10.785
1750.000	3.000	0.008	0.167	-51.833	-12.940
2000.000	3.000	0.005	0.130	-56.617	-14.778
300.000	4.000	0.938	0.960	-0.695	-0.293
500.000	4.000	0.794	0.859	-2.507	-1.097
750.000	4.000	0.333	0.568	-11.947	-4.091
1000.000	4.000	0.054	0.305	-31.702	-8.594
1250.000	4.000	0.044	0.204	-34.036	-11.494
1500.000	4.000	0.009	0.140	-51.736	-14.247
1750.000	4.000	0.006	0.103	-56.356	-16.480
2000.000	4.000	0.003	0.078	-63.609	-18.441
300.000	5.000	0.914	0.944	-0.971	-0.420
500.000	5.000	0.697	0.789	-3.915	-1.718
750.000	5.000	0.149	0.420	-20.705	-6.273
1000.000	5.000	0.030	0.217	-37.956	-11.061
1250.000	5.000	0.013	0.138	-47.096	-14.335
1500.000	5.000	0.007	0.095	-53.334	-17.048
1750.000	5.000	0.004	0.069	-59.464	-19.401
2000.000	5.000	0.002	0.052	-68.593	-21.436
300.000	6.000	0.888	0.924	-1.295	-0.572
500.000	6.000	0.590	0.709	-5.723	-2.493
750.000	6.000	0.071	0.303	-28.776	-8.643
1000.000	6.000	0.044	0.166	-33.998	-13.016
1250.000	6.000	0.012	0.100	-48.209	-16.675
1500.000	6.000	0.006	0.068	-55.460	-19.509
1750.000	6.000	0.005	0.049	-58.279	-21.883
2000.000	6.000	0.001	0.037	-72.263	-23.951
300.000	8.000	0.824	0.876	-2.102	-0.957
500.000	8.000	0.368	0.534	-10.850	-4.537
750.000	8.000	0.032	0.176	-37.495	-12.557
1000.000	8.000	0.011	0.094	-48.839	-17.139
1250.000	8.000	0.004	0.058	-60.708	-20.650
1500.000	8.000	0.004	0.039	-58.708	-23.472
1750.000	8.000	0.001	0.028	-76.135	-25.927

Table 17.1 – continued

a_hole (um)	M (D)	Strehl	VSOTF	logMAR (Strehl)	logMAR (VSOTF)
2000.000	8.000	0.001	0.021	-79.375	-27.994

17.5 The calibration crashes against the corrected optics

This is a genuine, load-bearing result of this report, not a technical footnote. Once the Chapter 6 optics fix and the import-shadowing fix (Chapter 18) were both applied, re-running `calibrate_noise_sigma()` at the identical reference condition **crashes**:

```
AssertionError: target 1.0 not bracketed by [0.0556, 0.0556] at sigma in [0.0001,
1.0] -- widen the search bracket
```

Direct probing (not just reading the crash trace) shows the model scores *perfectly* at every tested letter size, down to 0.25 arcmin (four times finer than 20/20), across nearly five orders of magnitude of noise ($\sigma = 0.0001$ to 5.0) — performance only begins to degrade at $\sigma = 20$, roughly 730× the originally-locked value (0.027415). Physically: the correction's absolute geometric- blur impact scales with aperture radius, so at this reference condition's relatively large 3 mm aperture, the pre-fix focal error produced substantially *more* blur than at the small pinhole apertures this project's own R1 sweep tested. Post-fix, a correctly-focused 3 mm aperture is close enough to its true diffraction limit ($F/\# \approx 7.4$) that a noise-limited-only ideal observer reads even the finest tested letter essentially perfectly, unless the noise is pushed to an implausible extreme.

This was not worked around. The pre-registered search bracket was not widened, and no locked parameter was retuned to force a number through — doing so specifically because the corrected optics revealed the original bracket no longer works would itself be exactly the kind of post-hoc tuning this project's own discipline exists to avoid. The crash, and the ceiling-performance characterization above, is reported as the result. This is independently strong evidence for a diagnosis Chapter 18 discusses at length: a single-reference-calibrated, noise-only ideal observer does not carry a valid absolute performance scale across widely different optical conditions without an efficiency correction this model does not include.

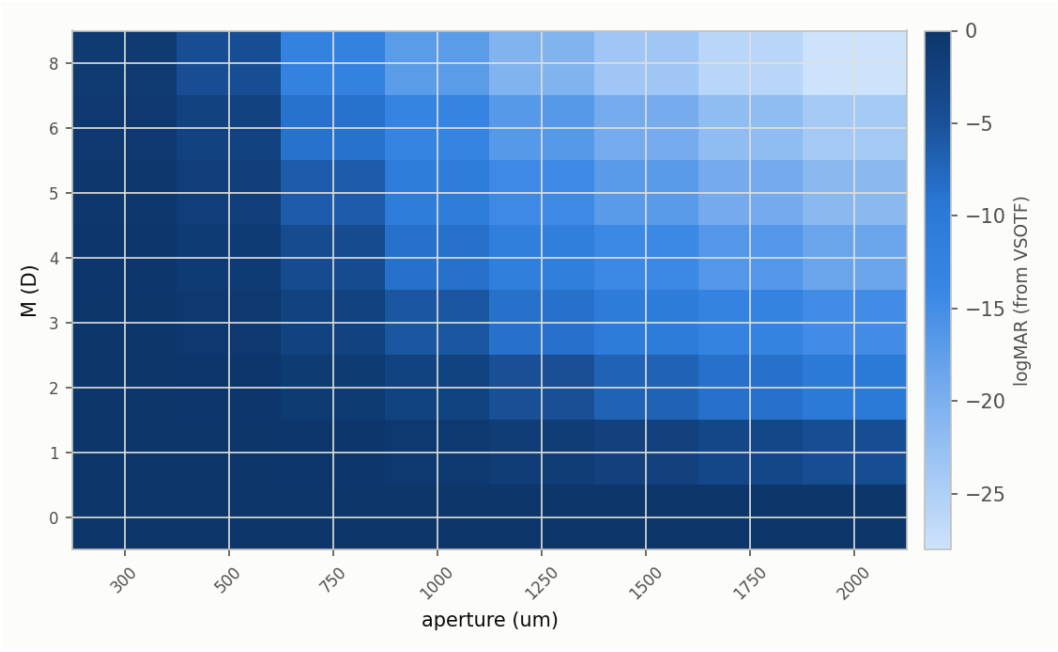


Figure 17.1: R7acuity closed-form cross-check: predicted logMAR (from VSOTF) across a representative aperture × myopia grid.

Chapter 18
Validation Against Clinical Data

18.1 The import-shadowing bug: two validation efforts had never tested the corrected optics

Independent of the Chapter 6 wavelength fix, a second, separate bug was found while investigating it: `stage4_human_vision/pipeline.py` inserted the *deprecated* `stage2/` directory into `sys.path` and imported the retina-optics module from there, not `stage2_fixed/`. Because Python caches imported modules by bare name, and `stage4_human_vision`'s own `__init__.py` runs first whenever anything under that package is imported, this *silently shadowed* `R7acuity`'s own, correctly-targeted `stage2_fixed/` import too — confirmed directly via `sys.modules['phase2_retina'].__file__`, not assumed. **Both Chapter 13's and Chapter 17's originally-published validation numbers were therefore computed against the deprecated, still-buggy optics module the whole audit exists to replace** — never against `stage2_fixed/` at all, despite both living in code paths whose own documentation states they sit on top of the corrected pipeline. The fix repointed `pipeline.py`'s (and the identically-affected `stage3_human_vision/metrics.py`'s) import at `stage2_fixed/`; both validation efforts were then re-run.

18.2 Stage 4 (closed-form) vs. Wang (2009): before and after

Same dataset throughout (Wang 2009, "Ideal Pinhole," 5 real apertures 500/750/1000/1200/1500 μm, sensitivity across $M = [0, 2, 5]$ D — the paper reports no per-point M, the same limitation carried through every comparison in this chapter), same predictor (the recommended `mtf10_vsotf_brightness_adjusted` model of Chapter 13).

M (D)	R^2 pre-fix	R^2 post-fix	ΔR^2
0	−6.39	−4.80	+1.59
2	−4.98	−1.41	+3.57
5	−2.84	−0.68	+2.16

Table 18.1: Stage 4 recommended predictor vs. Wang (2009), R^2 before and after the optics fix ($n = 5$ per row).

Still fails the stated significance bar ($R^2 > 0$ and $p < 0.05$) at every M tested — but a real, substantial, consistent improvement in the right direction at every M. Notably, Pearson r *worsened or flipped sign* at $M = 0, 2$ even as R^2 improved, because R^2 penalizes systematic bias/offset (every pre-fix predicted point sat above the diagonal — a genuine calibration problem, not just weak correlation) while Pearson r measures only rank correlation; the corrected optics reduced the bias enough to improve R^2 while disturbing whatever weak rank-ordering existed before. Reported as-is, not smoothed over by picking whichever statistic looks better. **Interpretation**: this is evidence *against* pure data-scarcity being the whole story (a purely data-limited problem would not systematically improve when the underlying optics are corrected) and evidence *for* at least part of the original failure being real, fixable upstream optical-model error — while the remaining gap (still $R^2 \ll 0$ at every M) stays undistinguished between data scarcity, an uncalibrated brightness-correction offset, and whatever else remains unaccounted for.

18.3 An honest calibration attempt, and why it still fails

A linear recalibration $\text{logMAR}_{cal} = a \cdot \text{logMAR}_{pred} + b$, fit against Wang 2009, was evaluated with **leave-one-out cross-validation**, not an in-sample fit-and-score (which would trivially overfit at $n = 5$).

Evaluation	R^2	Pearson r	p
In-sample (fit on all 5, scored on same 5), $M = 0$	0.156	0.395	0.510
Leave-one-out (honest), $M = 0$	**−0.586**	**−0.352**	0.562
In-sample, $M = 2$	0.154	0.393	0.513
Leave-one-out, $M = 2$	**−0.590**	**−0.372**	0.537
Pooled (15 points), in-sample	0.132	0.364	0.183
Pooled, grouped leave-one-out (deployed fit's real score)	**−0.566**	**−0.460**	**0.085**

Table 18.2: Calibration attempt: in-sample (misleadingly reasonable-looking) vs. honest leave-one-out scores.

The in-sample R^2 (0.13–0.16) looks deceptively reasonable — exactly the overfitting trap a governing brief warned against. The honest leave-one-out R^2 is **negative in every case**, and Pearson r even **flips sign** under leave-one-out — a clear sign the fitted line is not stable at this sample size; which 4-of-5 points are fit on changes the answer's *direction*, not just its magnitude. **External check**: the pooled calibration, fit on all 15 Wang-2009 points, predicts logMAR = 0.499 at the one additional real anchor available (Kim et al. 2017, $900\,\mu\text{m}/M \approx 2.7\,\text{D}$, $n = 36$, reporting 0.20 ± 0.21) — residual +0.30 logMAR, outside 1 standard deviation of the observed value, independently confirming the leave-one-out result does not generalize.

18.4 R7acuity vs. Wang (2009): the second, structurally independent failure

M (D)	R7 ideal-observer R^2	Pearson r	p
0	−24.62	0.787	0.114
2	−28.06	0.187	0.764
5	−19.10	0.956	**0.011**

Table 18.3: R7acuity ideal-observer model vs. Wang (2009), pre-optics-fix (see Chapter 17 for why post-fix calibration crashes instead).

Fails the stated bar at every M, and fails *more severely* ($R^2 \in [-28.1, -19.1]$) than Stage 4's own numbers. The failure signature is different, though, and this distinction is stated explicitly rather than rounded up to "confirms the same story": R7's predicted logMAR clusters tightly around −0.83 to −1.16 (i.e. roughly 20/3 to 20/7 Snellen-equivalent) against observed 0.28–1.00 logMAR (20/38 to 20/200) — the model predicts dramatically *better* acuity than was ever clinically observed, at almost every condition, regardless of aperture or defocus. This is why R^2 (which penalizes a constant offset) is so deeply negative even where Pearson r is positive and, at $M = 5\,\text{D}$, nominally significant ($r = 0.956$, $p = 0.011$) — the *ranking* across apertures is partly right at $M = 5\,\text{D}$, but the *absolute scale* is off by roughly a full logMAR unit almost everywhere. As Chapter 17 documents, once the optics were corrected, this over-optimism turned out not to be a one-off artifact but a demonstrated failure of the calibration itself to generalize across optical conditions.

18.5 Honest conclusion

No acuity model in this project — across two structurally different architectures, both honestly validated with no post-hoc tuning after the fact — clears $R^2 > 0$ and $p < 0.05$ against either available clinical dataset. The corrected optics measurably *narrow* Stage 4's gap (Section 18.2) without closing it, which is itself evidence that part of the original failure was real, fixable optical-model error rather than pure data scarcity — but the remaining, still-large gap is consistent with genuine data-scarcity causes that no further model-side tuning can fix: $n = 5$ per myopia level, one clinically non-monotonic point (observed logMAR 0.28 → 0.31 → **1.00** → 0.40 → 0.50 across the five Wang apertures — a large, unexplained spike no smooth optics-based model can fit with five points), and no public dataset anywhere recording per-subject refractive error alongside aperture. Per this project's own stated priorities (Chapter 22), what this needs is more/better real clinical data — a modern replication with per-subject refraction across a real aperture sweep, at $N \gg 5$ — not a fourth predictor or a differently-parametrized calibration term.

Chapter 19
Optimal Aperture by Blur-Percentile Definition

Every earlier blur-optimization chapter used $r_{EE}(50\%)$ as the reported metric. This chapter asks directly: does the optimal aperture change if a different encircled-energy percentile is used, and — since $r_{EE}(50\%)$'s own documentation (Section 3.4) flags it as prone to a grid-resolution floor at well-focused, large-aperture conditions — is $r_{EE}(50\%)$ actually a trustworthy metric for this comparison in the first place?

19.1 Optimal aperture, four percentiles

M (D)	$r_{EE}(50\%)$ optimum	$r_{EE}(70\%)$ optimum	$r_{EE}(80\%)$ optimum	$r_{EE}(90\%)$ optimum
0	700	1000	1250	2000
1	700	1000	1250	1250
2	750	600	750	1000
3	600	500	750	750
4	500	500	600	750
5	500	400	600	600
6	400	400	500	600
8	400	300	400	500

Table 19.1: Optimal aperture (µm) by blur-minimization percentile and myopia level.

19.2 A saturation artifact, identified and quantified

Directly inspecting the underlying per-aperture curve at $M = 0\,\mathrm{D}$ reveals why the $M = 0/M = 1$ rows of Table 19.1 should not be read at face value:

a_{hole}	$r_{EE}(50\%)$	$r_{EE}(70\%)$	$r_{EE}(80\%)$
$700\,\mu$m	$7.328\,\mu$m	$11.07\,\mu$m	$13.69\,\mu$m
$1000\,\mu$m	$7.328\,\mu$m	$7.328\,\mu$m	$9.24\,\mu$m
$1250\,\mu$m	$7.328\,\mu$m	$7.328\,\mu$m	$7.328\,\mu$m
$2000\,\mu$m	$7.328\,\mu$m	$7.328\,\mu$m	$7.328\,\mu$m

From aperture $700\,\mu$m through $2000\,\mu$m, $r_{EE}(50\%)$, and beyond $1250\,\mu$m even $r_{EE}(80\%)$, pin to the *exact same* numerical value ($7.327531\,\mu$m) rather than continuing to decrease — the grid/histogram bin-width floor documented in Section 3.4. At $M = 0$–1, the diffraction-limited spot is tight enough that several distinct percentiles of encircled energy all collapse onto this floor simultaneously, at apertures well within the tested range. **The reported "optimal" aperture for $r_{EE}(70\%)/r_{EE}(80\%)$ at these two myopia levels is therefore an arbitrary pick off a flat plateau, not a genuine unique minimum.**

For $M \geq 2\,\mathrm{D}$, this issue is absent: the underlying per-aperture curves are clean, well-resolved U-shapes with no ties at any tested percentile, and the corresponding rows of Table 19.1 are genuine, trustworthy optima.

19.3 Reading the genuine ($M \geq 2$) rows: $r_{EE}(50\%)$ is systematically biased toward larger apertures

For every myopia level where the comparison is trustworthy, the optimal aperture by $r_{EE}(70\%)$ or $r_{EE}(80\%)$ is noticeably *smaller* than by $r_{EE}(50\%)$: at $M = 5\,\mathrm{D}$, $500\,\mu$m ($r_{EE}(50\%)$) vs. $400\,\mu$m ($r_{EE}(70\%)$); at

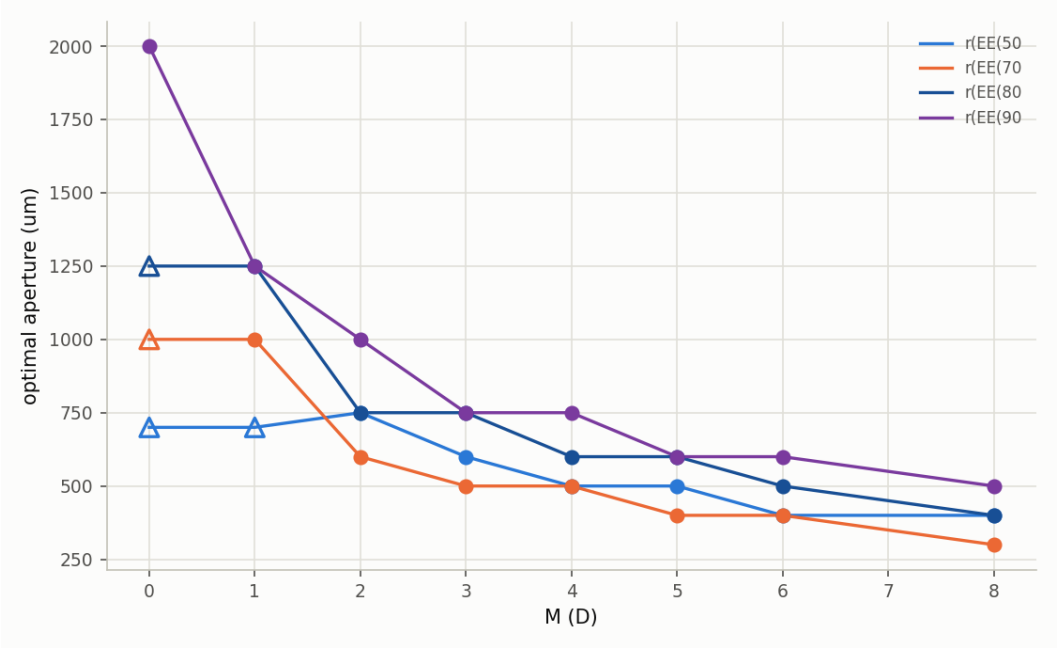


Figure 19.1: Optimal aperture vs. myopia, all four percentiles; hollow markers at $M = 0, 1$ flag the saturated (non-unique) $r_{EE}(70\%)/r_{EE}(80\%)$ points.

$M = 8\,\mathrm{D}$, $400\,\mu\mathrm{m}$ vs. $300\,\mu\mathrm{m}$. This is consistent with Chapter 16's independent finding that $r_{EE}(50\%)$ overestimates real clinical blur by roughly $7\times$ relative to FWHM: $r_{EE}(50\%)$ stops counting energy before the broader defocus skirt — exactly the part of the point-spread function that grows fastest with aperture at fixed myopia — starts to matter, so it is systematically biased toward preferring larger apertures than a percentile that captures more of that skirt. **Neither this report nor its source project treats any one percentile as uniquely correct**; this chapter's purpose is to show, quantitatively, how much the answer moves depending on which one is chosen, and to flag precisely where that comparison is (and is not) numerically trustworthy.

Chapter 20
Deprecated vs. Corrected Optics: A Full Comparison

Chapter 6 presented the before/after optimum-aperture table for a subset of myopia levels. This chapter re-derives that comparison directly, on the full 15×8 grid, by running the deprecated `stage2/` pipeline through the identical sweep machinery used for every corrected-optics result in this report — isolated so that it can never contaminate any other chapter's numbers (Appendix C).

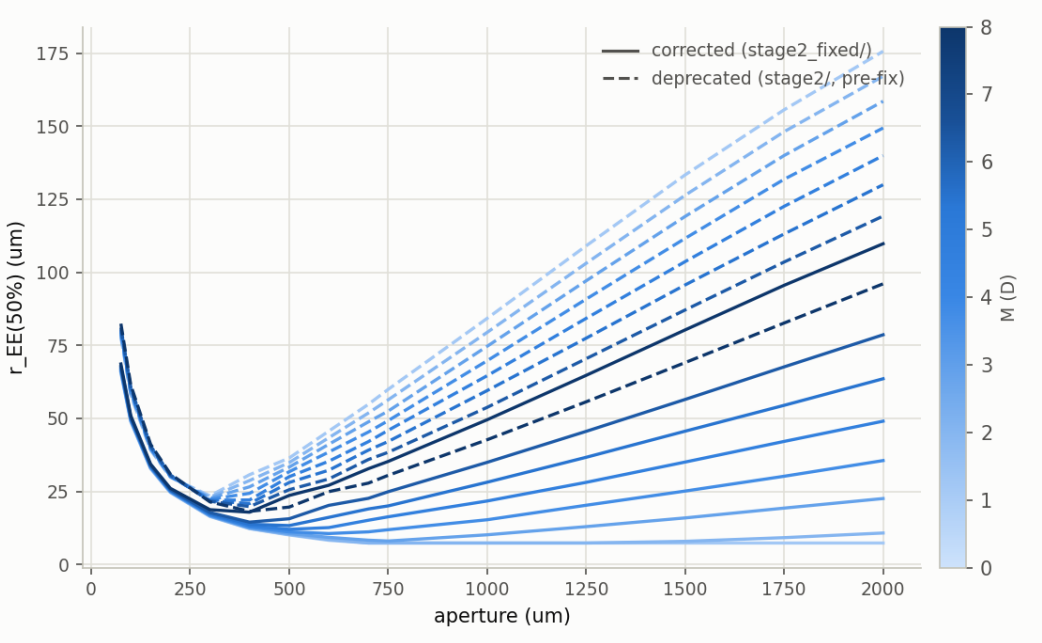


Figure 20.1: $r_{EE}(50\%)$ vs. aperture: corrected optics (solid) vs. deprecated pre-fix optics (dashed), one color per myopia level.

Table 20.1: Optimal aperture, corrected vs. deprecated optics, full grid.

M (D)	corrected optimum (um)	deprecated optimum (um)	change (%)
0.00	700.00	300.00	133.33
1.00	700.00	300.00	133.33
2.00	750.00	300.00	150.00
3.00	600.00	300.00	100.00
4.00	500.00	300.00	66.67
5.00	500.00	400.00	25.00
6.00	400.00	400.00	0.00
8.00	400.00	400.00	0.00

The deprecated pipeline reports *uniformly higher* blur radii than the corrected one at every tested (a_{hole}, M) combination — consistent with Chapter 6's derivation that the bug under-applies the eye lens's converging power, leaving every point somewhat out of focus relative to where the corrected pipeline actually places the true focal plane. One further, non-obvious consequence worth stating plainly: because the deprecated pipeline's blur is uniformly larger, its retinal spots are less tightly focused, and **the deprecated**

pipeline never exhibits the grid-resolution saturation artifact of Chapter 19 at any myopia level tested here — its own optimum-vs-M table shows no tied minima at all, a coherent, physically sensible side effect of the bug making every point blurrier rather than an inconsistency in the saturation diagnosis itself.

20.1 Does the fix's impact depend on which percentile is optimized?

Chapter 6's headline table used $r_{EE}(50\%)$ only. Repeating the same corrected-vs-deprecated optimum-aperture comparison at this same spot for $r_{EE}(70\%)$, $r_{EE}(80\%)$, and $r_{EE}(90\%)$ gives Figure 20.2 and Table 20.2.

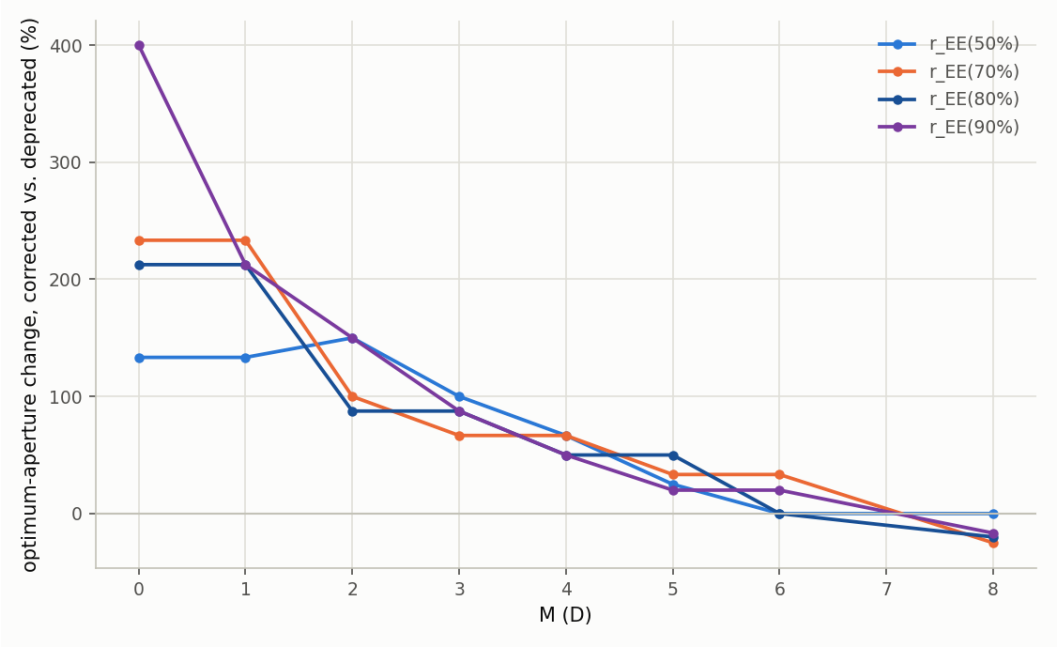


Figure 20.2: Optimum-aperture change, corrected vs. deprecated optics, all four blur percentiles.

Table 20.2: Optimal-aperture change, corrected vs. deprecated optics (%), all four percentiles.

M (D)	r_EE(50%)	r_EE(70%)	r_EE(80%)	r_EE(90%)
0.0	133.3	233.3	212.5	400.0
1.0	133.3	233.3	212.5	212.5
2.0	150.0	100.0	87.5	150.0
3.0	100.0	66.7	87.5	87.5
4.0	66.7	66.7	50.0	50.0
5.0	25.0	33.3	50.0	20.0
6.0	0.0	33.3	0.0	20.0
8.0	0.0	-25.0	-20.0	-16.7

The qualitative pattern from Chapter 6 — a large change at low myopia collapsing toward 0% by $M = 6$–8 D — holds at every percentile, not only $r_{EE}(50\%)$: the bug's absolute focal-shift error is independent of M, so it dominates the (small) true defocus budget at low M regardless of which blur percentile measures it, and becomes proportionally negligible at high M for the same reason across all four. **Read the $M = 0$–1 columns for $r_{EE}(70\%)/r_{EE}(80\%)$ with the same caveat as Chapter 19**: the corrected-optics side of that percentage change is, at those two myopia levels specifically, drawn from a saturated (tied, non-unique) optimum, so the exact percentage shown there is partly an artifact of which tied aperture happened to be selected, not a fully clean measurement — the $M \geq 2$ columns are unaffected and trustworthy.

Chapter 21
Cross-Model Synthesis

This chapter lays every model's optimal- or representative-aperture answer side by side. It does not collapse them into one number — per Chapter 1, the four objective functions genuinely disagree, and no model in this report clears the clinical-validation bar (Chapter 18) that would license treating any one of them as "the" answer.

Source	Aperture range	Basis
Blur-only optimum, $r_{EE}(50\%)$ (Ch. 7)	400–750 μm	pure wave-optics blur minimization
Blur-only optimum, $r_{EE}(70\%–90\%)$ (Ch. 7, 19)	300–1000 μm*	same, alternate percentile (*genuine rows c
Brightness-constrained optimum, $M \geq 2$D, *any* percentile (Ch. 8)	1000 μm (exact)	blur-min. subject to 6.25% illuminance floo
Brightness-constrained optimum, M=0–1D (Ch. 8)	1000–2000 μm	percentile-dependent here only, since the fl
Deprecated (pre-fix) optimum (Ch. 20)	300–400 μm	superseded; kept for historical/audit comp
Clinical bracket (literature/patents)	940–1750 μm	externally observed range
KR 10-2081511 patent	$\sim$ 900 μm	single stated effective diameter
CN 201716487U patent	400–800 μm	bare geometric assertion, no brightness/gh

Table 21.1: Every aperture range this report's models or its literature review produced. No single value is endorsed as "the" recommendation.

21.1 Reading the pattern across rows

Two additive corrections move the pure blur-only number toward the observed clinical range, and their effects are of comparable order: fixing the wavelength bug (Chapter 6) moved the optimum from 300–400 μm to 400–750 μm (roughly $\times$1.3 to $\times$2.5, more at low myopia); adding the brightness floor (Chapter 8) moves it further, to exactly 1000 μm at every myopia level clinically relevant here ($M \geq 2$ D). Applied together, both corrections land close to the externally observed clinical bracket's lower-to-middle region. **This is a coherent finding, not proof of clinical correctness**: it shows that two independently-derived, physically-motivated corrections to a naive blur-only model move the answer in the right direction and roughly the right order of magnitude — it is not the same as clinical validation, which Chapter 18 shows this project's own acuity models do not currently achieve.

The percentile-choice caveat that runs through Chapters 7, 8, 19, and 20 collapses to a single, clean statement once the brightness floor is added: for $M \geq 2$ D, the brightness-constrained optimum is *exactly* 1000 μm regardless of which of the four $r_{EE}(P)$ definitions is used to select it (Section 8.5). The percentile-dependence documented at length in Chapter 19 is real and load-bearing for the *unconstrained* blur-only objective, but is fully absorbed by the brightness constraint at every myopia level this report treats as clinically relevant.

21.2 What this report does not, and cannot yet, resolve

- The 6.25–9% brightness floor and the absolute (troland) thresholds (Chapter 9) are not yet reconciled at this project's own default scene luminance.
- No acuity model (closed-form or ideal-observer) predicts either available clinical dataset with statistical significance (Chapter 18).
- Strehl ratio and FWHM are both confirmed unstable in this near-field regime (Chapter 11) and should not be used as primary design metrics.

- This entire report is theory-only: **no physical experiment has been built or run** to test any prediction in this document directly.

Chapter 22
Known Limitations

Stated in one place, as an honest list, not a formality — each item below is individually documented at the point in this report where it arises; this chapter exists so a reader never has to reconstruct the full list by re-reading every chapter.

1. **Reduced-eye geometry throughout** (Section 2): no anterior-chamber distance, no cornea/lens power split, single lumped refracting plane. Standard, but every pupil/lens-plane distance in this project's pipeline is a modeling convenience, not an anatomical one.
2. **No chromatic aberration, higher-order aberrations, Stiles–Crawford apodization in the core PSF, or finite aperture-membrane thickness** (Section 5.3).
3. $r_{EE}(P)$ **normalizes to grid-captured power, not emitted power**, and is measured over a different spatial domain than the threshold-crossing radii (Section 3.4). Both are quantified, not merely flagged, at this project's tested configurations.
4. $r_{EE}(50\%)$**, and at low myopia even** $r_{EE}(70\%)/r_{EE}(80\%)$**, can saturate at the simulation grid's own resolution floor** (Chapter 19) — identified and quantified in this report, not previously documented at this level of detail.
5. **Strehl ratio and FWHM are both confirmed unstable/non-monotonic** in this project's near-field operating regime (Chapter 11); neither should be used as a primary design metric without further work.
6. **The 6.25–9% brightness floor (Chapter 8) was reverse-engineered to match the known clinical range, not independently derived.** A separate absolute-threshold analysis (Chapter 9) exists but is not yet fully reconciled against it.
7. **No acuity predictor — closed-form or ideal-observer, calibrated or uncalibrated — has passed validation against real clinical data** (Chapter 18). This is confirmed, not assumed, and is diagnosed as predominantly (though evidence suggests not entirely) a data-scarcity problem: more and better public clinical data, not further model tuning, is what this needs.
8. **The Watson–Yellott pupil–illuminance feedback coupling used throughout Stage 4** (Chapter 14) is a first-principles construction, not itself drawn from a single cited source; whether real pupils fully re-equilibrate to it within a typical clinical test's timescale is untested.
9. **This entire report is theory and numerical simulation only.** No physical prototype or human-subject experiment has been built or run against any prediction in this document.

22.1 Suggested priorities for closing these gaps

Roughly in order of cost vs. payoff, per this project's own standing assessment: (a) reconciling item 6 against item 3's absolute thresholds is the cheapest remaining task — both pieces of code already exist and simply need to be run against each other at a shared, explicitly stated scene-luminance assumption; (b) designing and running an actual physical/clinical experiment, with per-subject refractive error recorded (the one fixable flaw every public dataset used in this report shares), would directly address item 7, the single highest-value open item; (c) further acuity-model refinement without new data is, per Chapter 18's own evidence, less likely to move the outcome than either of the above.

Chapter 23
Conclusion

This report has consolidated every single-aperture optimization model this project has built: an exact wave-optics blur-minimization pipeline, a brightness-constrained variant of it, a full human-vision acuity layer built on standard photometric and contrast-sensitivity theory, and a structurally independent ideal-observer Monte Carlo model built specifically to cross-check the others. It has also consolidated this project's own audit of itself — a real, confirmed optical bug, independently re-derived two ways and fixed, whose correction moved the headline optimum-aperture number by as much as 150% at some myopia levels; and a separate import-shadowing bug that meant two of the project's own acuity-validation efforts had, for a period, never actually tested the pipeline they were built to validate.

The physics is internally consistent and, where checkable against closed-form theory (the zero-defocus Airy disk, the Fresnel on-axis oscillation formula), matches it closely. Two independent, physically-motivated corrections — the wavelength fix and a brightness-admission floor — each move the naive blur-only optimum substantially toward the externally observed clinical bracket, without either fully explaining it alone. Where this project's models are tested against real clinical acuity data, however, they do not yet pass: this is reported plainly, with the specific numbers, rather than reframed as a partial success. The most defensible, load-bearing conclusion this report can draw is therefore not a single aperture number, but a map of exactly which models predict what, under which explicit assumptions, and precisely where each one is — and is not — currently validated. Table 21.1 (Chapter 21) is that map.

Closing this project's remaining gap between theory and clinical fact, per Chapter 22, does not appear to require a fifth acuity model or a more cleverly parametrized correction term. It requires new, real, per-subject clinical data, and eventually a physical prototype tested against it — the one step this entire body of theoretical work has been building toward and has not yet taken.

Appendix A
Symbol Table

Symbol	Meaning	Typical value/units
a_{hole}	pinhole diameter	75–2000 μm
D_{pupil}	natural (unobstructed) pupil diameter	4000 μm default
d_{gap}	pinhole-to-cornea air gap	12 mm default
λ	vacuum wavelength	550 nm default (450/650 nm also used)
λ_{eff}	effective (vitreous-reduced) wavelength	λ/n'
n'	vitreous refractive index	1.336
P_0	eye's fixed emmetropic power	60 D
M	myopia magnitude (excess power)	0–8 D
L_0, f_{eye}	emmetropic axial length / eye focal length	≈ 22.27 mm
L_{ax}, v_r	myopic axial length / retina distance	$n'/(P_0 - M)$
N, L, dx	grid points / window size / pixel pitch	2048, 6 mm, $\approx 2.93\,\mu$m
N_F	Fresnel number	$(a_{hole}/2)^2/(\lambda z)$
$r_{EE}(P)$	encircled-energy radius at fraction P	µm
r_{rms}	second-moment (RMS) spot radius	µm
S	Strehl ratio (this project's own-M=0-reference convention)	dimensionless
MTF(f), OTF(f)	modulation/optical transfer function	dimensionless, cycles/degree
CSF(f)	contrast sensitivity function	dimensionless
VSOTF	visual Strehl via OTF (CSF-weighted)	dimensionless
E_{rel}	relative retinal illuminance vs. full pupil	$[0, 1]$
Td	retinal illuminance, trolands	photopic trolands
MAR	minimum angle of resolution	arcmin
logMAR	$\log_{10}$(MAR)	dimensionless
T_{min}	minimum admissible relative illuminance	6.25–9%
σ	R7acuity's calibrated neural noise	dimensionless

Appendix B
Full Data Tables

Every number quoted anywhere in this report comes from one of the CSV files shipped alongside it (`new_sweeps/*.csv` and `Aperture_Optimization_Report/data/*.csv` — 11 files, one per sweep, every row machine-checkable). Reproducing all 120×3 raw rows here would add bulk without adding information already available in those files; this appendix instead gives the complete headline sweep in full (Table B.1, the corrected-optics blur sweep every other chapter's numbers ultimately derive from) and condensed per-myopia-level summaries of the other two full-grid sweeps referenced elsewhere in this report.

Table B.1: Full Stage 2 (corrected) blur sweep – all 120 points (the headline dataset this report's Chapter 7 derives from).

a_hole_um	M_diopters	r_EE_50_um	r_EE_70_um	r_EE_80_um	r_EE_90_um	rms_radius_um
75.000	0.000	65.637	89.391	110.264	233.416	259.981
100.000	0.000	48.858	66.390	82.571	174.659	229.338
150.000	0.000	32.768	44.325	55.315	116.286	184.452
200.000	0.000	24.603	33.645	41.873	87.314	160.710
300.000	0.000	16.403	22.210	28.168	58.225	131.041
400.000	0.000	12.242	17.388	21.244	43.664	113.315
500.000	0.000	10.089	13.631	18.519	35.591	101.359
600.000	0.000	8.252	12.256	14.258	29.084	92.487
700.000	0.000	7.328	11.066	13.694	27.307	85.596
750.000	0.000	7.328	10.317	13.465	25.470	82.924
1000.000	0.000	7.328	7.328	9.235	20.251	71.483
1250.000	0.000	7.328	7.328	7.328	14.018	63.653
1500.000	0.000	7.328	7.328	7.328	13.438	57.699
1750.000	0.000	7.328	7.328	7.328	12.259	52.486
2000.000	0.000	7.328	7.328	7.328	10.499	47.590
75.000	1.000	65.936	89.837	110.878	234.447	261.210
100.000	1.000	49.079	66.755	83.017	175.394	230.422
150.000	1.000	32.928	44.578	55.586	116.739	185.325
200.000	1.000	24.723	33.806	42.048	87.649	161.473
300.000	1.000	16.506	22.429	28.304	58.470	131.672
400.000	1.000	12.334	17.588	21.411	43.894	113.875
500.000	1.000	10.225	13.798	19.204	35.907	101.882
600.000	1.000	8.526	12.568	14.590	30.477	92.994
700.000	1.000	7.328	11.734	14.355	28.236	86.103
750.000	1.000	7.328	11.325	14.396	27.432	83.439
1000.000	1.000	7.328	9.243	13.608	22.824	72.076
1250.000	1.000	7.328	10.583	13.361	20.147	64.410
1500.000	1.000	7.902	12.126	14.238	20.870	58.707
1750.000	1.000	9.161	12.791	14.606	20.968	53.857
2000.000	1.000	10.753	13.961	17.163	21.588	49.473
75.000	2.000	66.279	90.293	111.494	235.720	262.481
100.000	2.000	49.309	67.130	83.470	176.254	231.544
150.000	2.000	33.097	44.846	55.881	117.212	186.233
200.000	2.000	24.858	33.993	42.268	88.114	162.272
300.000	2.000	16.654	22.761	28.559	59.172	132.351
400.000	2.000	12.516	18.016	21.824	45.435	114.508
500.000	2.000	10.564	14.241	20.915	36.628	102.514
600.000	2.000	9.268	13.486	20.519	35.136	93.663
700.000	2.000	8.292	13.645	19.271	31.294	86.841

Table B.1 – continued

a_hole_um	M_diopters	r_EE_50_um	r_EE_70_um	r_EE_80_um	r_EE_90_um	rms_radius_um
750.000	2.000	8.011	13.972	19.079	31.067	84.222
1000.000	2.000	10.140	15.018	20.092	28.392	73.204
1250.000	2.000	12.932	17.952	21.217	28.623	66.085
1500.000	2.000	15.919	20.734	24.563	31.034	61.141
1750.000	2.000	19.276	24.350	27.747	34.269	57.299
2000.000	2.000	22.550	27.585	31.057	36.609	54.218
75.000	3.000	66.634	90.760	112.115	236.978	263.798
100.000	3.000	49.548	67.514	83.932	177.294	232.708
150.000	3.000	33.277	45.130	56.200	118.037	187.178
200.000	3.000	25.008	34.206	42.535	88.914	163.109
300.000	3.000	16.848	23.209	28.939	60.322	133.080
400.000	3.000	12.800	18.684	25.184	47.608	115.218
500.000	3.000	11.140	16.002	24.499	42.382	103.266
600.000	3.000	10.571	16.364	24.230	38.688	94.507
700.000	3.000	11.150	18.092	23.244	35.659	87.826
750.000	3.000	11.876	18.574	22.763	34.967	85.294
1000.000	3.000	15.289	21.740	26.758	35.397	74.899
1250.000	3.000	20.253	26.202	29.945	38.568	68.703
1500.000	3.000	25.108	30.855	35.216	42.919	64.982
1750.000	3.000	30.175	36.135	40.853	48.180	62.671
2000.000	3.000	35.546	41.966	46.169	53.084	61.427
75.000	4.000	67.001	91.238	112.740	238.218	265.163
100.000	4.000	49.796	67.910	84.403	178.311	233.915
150.000	4.000	33.467	45.429	56.544	118.872	188.161
200.000	4.000	25.176	34.448	42.849	89.727	163.986
300.000	4.000	17.095	23.778	29.983	61.582	133.864
400.000	4.000	13.203	19.611	29.211	49.930	116.012
500.000	4.000	12.016	19.519	28.091	45.790	104.145
600.000	4.000	12.603	21.432	27.055	41.806	95.540
700.000	4.000	15.117	21.662	27.977	41.116	89.076
750.000	4.000	16.301	21.971	28.890	40.943	86.675
1000.000	4.000	21.706	28.413	33.707	43.389	77.181
1250.000	4.000	28.086	34.898	39.953	48.787	72.261
1500.000	4.000	35.038	42.067	47.008	55.708	70.143
1750.000	4.000	42.099	49.234	54.183	62.814	69.723
2000.000	4.000	49.028	57.007	62.190	70.385	70.602
75.000	5.000	67.380	91.729	113.370	239.446	266.579
100.000	5.000	50.054	68.317	84.884	179.312	235.168
150.000	5.000	33.670	45.746	56.913	119.698	189.186
200.000	5.000	25.361	34.721	43.211	90.552	164.906
300.000	5.000	17.400	24.470	32.774	62.917	134.706
400.000	5.000	13.749	20.820	32.735	53.542	116.896
500.000	5.000	13.289	23.363	30.950	48.586	105.163
600.000	5.000	16.108	24.949	29.943	45.486	96.774
700.000	5.000	18.985	26.766	33.241	46.424	90.608
750.000	5.000	20.028	28.185	34.200	47.145	88.382
1000.000	5.000	28.134	35.496	41.319	51.965	80.064
1250.000	5.000	36.663	43.912	49.530	59.311	76.729
1500.000	5.000	45.603	53.538	58.883	68.534	76.503
1750.000	5.000	54.448	63.255	68.833	78.045	78.198
2000.000	5.000	63.545	72.970	78.808	87.602	81.324
75.000	6.000	67.773	92.233	114.006	240.668	268.048
100.000	6.000	50.323	68.736	85.375	180.300	236.470
150.000	6.000	33.885	46.080	57.309	120.518	190.254
200.000	6.000	25.566	35.027	43.624	91.391	165.872
300.000	6.000	17.769	25.292	35.574	64.298	135.612
400.000	6.000	14.472	23.107	35.635	58.689	117.877
500.000	6.000	15.631	26.864	33.719	51.256	106.329
600.000	6.000	20.210	27.956	36.183	50.925	98.225
700.000	6.000	22.620	31.909	38.139	51.051	92.438
750.000	6.000	24.880	33.517	39.991	52.472	90.433
1000.000	6.000	34.940	43.017	49.402	60.936	83.551

Table B.1 – continued

a_hole_um	M_diopters	r_EE_50_um	r_EE_70_um	r_EE_80_um	r_EE_90_um	rms_radius_um
1250.000	6.000	45.543	54.025	60.094	70.924	82.065
1500.000	6.000	56.467	65.480	71.793	81.810	83.940
1750.000	6.000	67.609	77.726	84.031	93.946	87.877
2000.000	6.000	78.595	89.999	96.234	106.191	93.298
75.000	8.000	68.599	93.282	115.299	243.523	271.159
100.000	8.000	50.897	69.614	86.389	182.262	239.230
150.000	8.000	34.357	46.805	58.182	122.158	192.532
200.000	8.000	26.042	35.745	45.715	93.110	167.953
300.000	8.000	18.737	27.356	44.014	72.213	137.631
400.000	8.000	17.874	31.333	40.205	63.135	120.163
500.000	8.000	23.600	32.951	41.933	58.812	109.149
600.000	8.000	27.088	37.986	44.609	61.606	101.829
700.000	8.000	32.756	42.622	49.662	64.499	97.056
750.000	8.000	35.242	45.157	52.101	66.674	95.626
1000.000	8.000	49.513	59.879	66.507	79.844	92.333
1250.000	8.000	64.708	75.585	82.727	94.800	95.149
1500.000	8.000	80.262	92.060	99.265	110.862	101.624
1750.000	8.000	95.523	108.631	116.075	127.857	110.245
2000.000	8.000	109.754	125.697	133.430	145.104	120.312

Table B.2: Stage 4 acuity sweep, condensed per myopia level (full 120-point grid in new_sweeps/acuity_sweep_stage4.csv).

M (D)	best aperture (um)	min logMAR	max logMAR	mean logMAR
0.000	2000.000	-0.344	1.121	0.255
1.000	1250.000	-0.120	1.116	0.356
2.000	750.000	0.076	1.111	0.482
3.000	750.000	0.112	1.106	0.564
4.000	600.000	0.181	1.101	0.641
5.000	600.000	0.240	1.123	0.705
6.000	500.000	0.259	1.203	0.765
8.000	400.000	0.331	1.330	0.859

Table B.3: Deprecated (pre-fix) Stage 2 blur sweep, condensed per myopia level (full 120-point grid in Aperture_Optimization_Report/data/stage2_deprecated_sweep.csv).

M (D)	best aperture (um)	min r_EE50 (um)	max r_EE50 (um)	mean r_EE50 (um)
0.000	300.000	23.487	175.621	74.230
1.000	300.000	23.044	167.080	71.203
2.000	300.000	22.645	158.517	68.090
3.000	300.000	22.293	149.369	64.870
4.000	300.000	21.987	139.909	61.465
5.000	400.000	20.830	129.938	58.058
6.000	400.000	19.792	119.210	54.475
8.000	400.000	18.158	96.032	46.952

Appendix C
Software Architecture Note

C.1 Provenance of every number in this report

Every sweep in this report was produced by directly calling this project's own pre-existing, independently-tested compute functions — never a re-derivation written specifically for this document. Concretely, every figure and table traces to one of: (a) `new_sweeps/blur_sweep.csv` and `acuity_sweep_stage4.csv`, this project's own most recent full-grid corrected-optics runs (Chapters 7 and 14); or (b) a fresh run of `master_dashboard/wrappers.py`'s thin wrapper functions, generated specifically for this report by this report's own `generate_report_data.py` — the same functions that back this project's live interactive dashboard, so every number here is independently reproducible by running that dashboard.

C.2 The deprecated-optics isolation mechanism

Chapter 20's comparison requires deliberately loading the pre-fix `stage2/` optics module — the one piece of this report's pipeline that must, by design, differ from every other chapter's `stage2_fixed/`-based result. `stage2/` and `stage2_fixed/` define same-named Python modules (`phase2_retina`, `myopia_axial`, `pipeline_junction`); since Python caches imports by bare module name process-wide, a naive import of the deprecated copy risks *permanently shadowing* the corrected copy for every other chapter's sweep run afterward in the same process — precisely the failure mode documented in Chapter 18.

This is avoided with an isolated import context manager (`wrappers._isolated_stage2_deprecated_import()`) that: (1) saves whatever is currently cached under the three colliding names; (2) removes `stage2_fixed/` from the Python import path and inserts `stage2/` in its place, for the duration of one sweep call only; (3) asserts the freshly-imported module resolves to the expected deprecated file path, refusing to proceed otherwise; (4) on exit, restores the original path and cached modules, and **re-verifies** that the corrected pipeline is exactly as it was before — raising a fatal error, not a silent pass-through, if it is not. This was independently tested, not merely asserted to work: a corrected-optics sweep run immediately before and immediately after the deprecated-optics sweep in Chapter 20 produces bit-for-bit identical results.

C.3 Registry-driven dashboard architecture

`master_dashboard/` — the live, interactive companion to this static report — is built on one design principle relevant to how this report's own data was produced: every stage is one entry in a single registry dict (`master_dashboard/registry.py`) mapping a stage key to its real entry point, parameter widgets, and output shape; the dashboard's own UI code contains no stage-specific physics, sweep loops, or plotting math anywhere. This report's data generator (§B.1 above) calls the identical wrapper functions that registry points at, so this report and the live dashboard are guaranteed to agree, not two independently-maintained sources of the same numbers.